

MUNBYN

ITPP047 Thermal Printer

Program Manual

1.00

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1.0. Overview

This manual applies to Munbyn's 80mm Series Thermal Receipt Printer POS-80XX series. Command description of the programming manual includes the following sections:

- 1) An overview of the command names and functions. This is the first part of the command description. ASCII code is given command of the form and function of the command Can be summarized.
- 2) Format. This section uses the ASCII coding forms, hexadecimal code form, decimal code in the form of three forms describing the life Order.
Which part of the range, compared with a decimal number without special instructions, such as in the following example $1 \leq n \leq 4$, where 1 is a decimal number 1, rather than the ASCII code table "1."
- 3) Range. Gives a range of variables.
- 4) Description. Gives a detailed explanation of the command.
- 5) Annotations. Notice is given command. Since the command in the different modes, with different commands, may cause the phase Mutual influence, and this section gives the details.
- 6) Reference. Associated with this command is given, similar to other commands.

2.0. Commands

2.1. HT

[Name] Horizontal tab

[Format] ASCII: HT

Hex: 09

Decimal: 9

[Description] Moves the print position to the next horizontal tab position.

[Details]

- This command is ignored unless the next horizontal tab position has been set.
- If the next horizontal tab position exceeds the printing area, the printer sets the printing position to [Printing area width + 1].
- Horizontal tab positions are set with ESC D.
- If this command is received when the printing position is at [printing area width + 1], the printer executes print buffer- full printing of the current line and horizontal tab processing from the beginning of the next line.
- The default setting of the horizontal tab position for the paper roll is font A (12 × 24) every 8th character (9th, 17th, 25th, column).

[Reference] ESC D

2.2. LF

[Name] Print and line feed

[Format] ASCII LF

Hex 0A

Decimal 10

[Description] Prints the data in the print buffer and feeds one line based on the current line spacing. [Note] This command sets the print position to the beginning of the line.

[Reference] ESC 2, ESC 3

2.3. FF(*)

[Name] Print and return to standard mode in page mode

[Format] ASCII FF

Hex 0C

Decimal 12

[Description] Prints the data in the print buffer collectively and returns to standard mode.

[Details]

- The buffer data is deleted after being printed.
- The printing area set by ESC W is reset to the default setting.
- The printer does not execute paper cutting.
- This command sets the print position to the beginning of the line.
- This command is enabled only in page mode.

[Reference] ESC FF, ESC L, ESC S

2.4. CR

[Name] Print and carriage return

[Format]	ASCII	CR
	Hex	0D
	Decimal	13

[Description] When automatic line feed is enabled, this command functions the same as LF; When automatic line feed is disabled, this command is ignored.

[Details]

- Sets the print starting position to the beginning of the line.
- The automatic line feed is ignored with a serial interface model.
- This command is set according to the DIP switch 1-1 setting with a parallel interface model.

[Reference] LF

2.5. CAN (*)

[Name] Cancel print data in page mode

[Format]	ASCII	CAN
	Hex	18
	Decimal	24

[Description] In page mode, deletes all the print data in the current printable area.

[Details]

- This command is enabled only in page mode.
- If data that existed in the previously specified printing area also exists in the currently specified printing area, it is deleted.

[Reference] ESC L, ESC W

2.6. DLE EOT n (*)

[Name] Real-time status transmission

[Format]	ASCII	DLE	EOT	n
	Hex	10	04	n
	Decimal	16	4	n

[Range] $1 \leq n \leq 4$

[Description] Transmits the selected printer status specified by n in real time, according to the following parameters:

n = 1: Transmit printer status

n = 2: Transmit off-line status

n = 3: Transmit error status

n = 4: Transmit paper roll sensor status

[Details]

- The status is transmitted whenever the data sequence of <10>H<04>H< n> ($1 \leq n \leq 4$) is received.
Example:
In ESC * m nL nH d1. dk, d1=<10>H, d2=<04>H, d3=<01>H
- This command should not be used within the data sequence of another command that consists of 2 or more bytes. Example:

If you attempt to transmit ESC 3 n to the printer, but DTR (DSR for the host computer) goes to MARK before n is transmitted and then DLE EOT 3 interrupts before n is received, the code <10>H for DLE EOT 3 is processed as the code for ESC 3 <10>H.

- Even though the printer is not selected using ESC = (select peripheral device), this command is effective.
- The printer transmits the current status. Each status is represented by one byte.
- The printer transmits the status without confirming whether the host computer can receive data.
- The printer executes this command upon receiving it.
- This command is executed even when the printer is off-line, the receive buffer is full, or there is an error status with a serial interface model.
- With a parallel interface model, this command cannot be executed when the printer is busy. This command is executed even when the printer is off-line or there is an error status when DIP switch 2-1 is on with a parallel interface model.
- When Auto Status Back (ASB) is enabled using the GS a command, the status in the DLE EOT command and the ASB status must be differentiated.

Table 2-1 n = 1: Printer Status

Bit	Off/On	Hex	Decimal	Function
0	0	00	0	Not used. Fixed to Off.
1	1	02	2	Not used. Fixed to On.
2	0	00	0	One or two cash drawer open
	1	04	4	Both of two cash drawer closed
3	0	00	0	On-line
	1	08	8	Off-line
4	1	10	16	Not used. Fixed to On.
5,6				Undefined.
7	0	00	00	Not used. Fixed to Off.

Table 2-2 n = 2: Off-line Status

Bit	Off/On	Hex	Decimal	Function
0	0	00	0	Not used. Fixed to Off.
1	1	02	2	Not used. Fixed to On.
2	0	00	0	Cover is closed.
	1	04	4	Cover is open.
3	0	00	0	Paper is not being fed by using the FEED button.
	1	08	8	Paper is begin fed by the FEED button.
4	1	10	16	Not used. Fixed to On.
5	0	00	0	No paper-end stop.

	1	20	32	Printing is being stopped.
6	0	00	0	No error.
	1	40	64	Error occurs.
7	0	00	0	Not used. Fixed to Off.

Table 2-3 n = 3: Error Status

Bit	Off/On	Hex	Decimal	Function
0	0	00	0	Not used. Fixed to Off.
1	1	02	2	Not used. Fixed to On.
2				Undefined.
3	0	00	0	No auto-cutter error.
	1	08	8	Auto-cutter error occurs.
4	1	10	16	Not used. Fixed to On.
5	0	00	0	No unrecoverable error.
	1	20	32	Unrecoverable error occurs.
6	0	00	0	No auto-recoverable error.
	1	40	64	Auto recoverable error occurs.
7	0	00	0	Not used. Fixed to Off.

Table 2-4 n = 4: Continuous paper sensor Status

Bit	Off/On	Hex	Decimal	Function
0	0	00	0	Not used. Fixed to Off.
1	1	02	2	Not used Fixed to On.
2, 3	0	00	0	Paper roll near-end sensor: paper adequate.
	1	0C	12	Paper near-end is detected by the paper roll near-end sensor.
4	1	10	16	Not used. Fixed to On.
5, 6	0	00	0	Paper roll sensor: Paper present.
	1	60	96	Paper roll end detected by paper roll senso.
7	0	00	0	Not used. Fixed to Off.

[Reference] DLE ENQ, GS a, GS r

2.7. DLE ENQ n (*)

[Name] Real-time request to printer

[Format] ASCII DLE ENQ n
 Hex 10 05 n
 Decimal 16 5 n

[Range] $1 \leq n \leq 2$

[Description] Responds to a request from the host computer. n specifies the requests as follows:

Table 2-5 Request

n	Request
1	Recover from an error and restart printing from the line where the error occurred
2	Recover from an error aft clearing the receive and print buffers

[Details]

- This command is effective only when an auto-cutter error occurs.
- The printer starts processing data upon receiving this command.
- This command is executed even when the printer is off-line, the receive buffer is full, or there is an error status with a serial interface model.
- With a parallel interface model, this command cannot be executed when the printer is busy. This command is executed even when the printer is off-line or there is an error status when DIP switch 2-1 is on with a parallel interface model.
- The status is also transmitted whenever the data sequence of <10>H<05>H<n> ($1 \leq n \leq 2$) is received.
 Example:
 In ESC * m nL nH dk, d1 = <10>H, d2 = <05>H, d3 = <01>H
- This command should not be contained within another command that consists of two or more bytes.
 Example: If you attempt to transmit ESC 3 n to the printer, but DTR (DSR for the host computer) goes to MARK before n is transmitted, and DLE ENQ 2 interrupts before n is received, the code <10>H for DLE ENQ 2 is processed as the code for ESC 3 <10>H.
- DLE ENQ 2 enables the printer to recover from an error after clearing the data in the receive buffer and the print buffer. The printer retains the settings (by ESC !, ESC 3, etc.) that were in effect when the error occurred. The printer can be initialized completely by using this command and ESC @. This command is enabled only for errors that have the possibility of recovery, except for print head temperature error.
- When the printer is disabled with ESC = (Select peripheral device), the error recovery functions (DLE ENQ 1 and DLE ENQ 2) are enabled, and the other functions are disabled.

[Reference] DLE EOT

2.8. DLE DC4 n m t (*)

[Name] Generate pulse at real-time

[Format] ASCII DLE DC4 n m t
 Hex 10 14 n m t
 Decimal 16 20 n m t

[Range] n = 1

m = 0, 1

$1 \leq t \leq 8$

[Description] Outputs the pulse specified by t to connector pin m as follows:

Table 2-6 Connector Pin

m	Connector pin
0	Drawer kick-out connector pin 2.
1	Drawer kick-out connector pin 5.

pulse ON time is [$t \times 100$ ms] and the OFF time is [$t \times 100$ ms].

[Details]

- When the printer is in an error status when this command is processed, this command is ignored.
- When the pulse is output to the connector pin specified while ESC p or DEL DC4 is executed while this command is processed, this command is ignored.
- The printer executes this command upon receiving it.
- With a serial interface model, this command is executed even when the printer is off-line, the receive buffer is full, or there is an error status.
- With a parallel interface model, this command cannot be executed when the printer is busy. This command is executed even when the printer is off-line or there is an error status when DIP switch 2-1 is on.
- If print data includes the same character strings as this command, the printer performs the same operation specified by this command. The user must consider this.
- This command should not be used within the data sequence of another command that consists of 2 or more bytes.
- This command is effective even when the printer is disabled with ESC = (Select peripheral device).

[Reference] ESC p

2.9. ESC FF (*)

[Name] Print data in page mode

[Format] ASCII ESC FF

Hex 1B 0C

Decimal 27 12

[Description] In page mode, prints all buffered data in the printing area collectively.

[Details]

- This command is enabled only in page mode.
- After printing, the printer does not clear the buffered data, setting values for ESC T and ESC W, and the position for buffering character data.

[Reference] FF, ESC L, ESC S

2.10. ESC SP n

[Name] Set right-side character spacing

[Format] ASCII ESC SP n

Hex 1B 20 n

Decimal 27 32 n

[Range] $0 \leq n \leq 255$

[Description] Sets the character spacing for the right side of the character to [n horizontal or vertical motion units].

[Details]

- The right-side character spacing for double-width mode is twice the normal value. When characters are enlarged, the right-side character spacing is n times normal value.
- This command does not affect the setting of kanji characters.
- This command sets values independently in each mode (standard and page modes).
- The horizontal and vertical motion units are specified by GS P. Changing the horizontal or vertical motion unit does not affect the current right-side spacing.
- The GS P command can change the horizontal (and vertical) motion unit. However, the value cannot be less than the minimum horizontal movement amount, and it must be in even units of the minimum horizontal movement amount.
- In standard mode, the horizontal motion unit is used.
- In page mode, the horizontal or vertical motion unit differs in page mode, depending on starting position of the printable area as follows:
- When the starting position is set to the upper left or lower right of the printable area using ESC T, the horizontal motion unit (x) is used.
- When the starting position is set to the upper right or lower left of the printable area using ESC T, the vertical motion unit (y) is used.
- The maximum right-side spacing is 255/180 inches. Any setting exceeding the maximum is converted to the maximum automatically.

[Default] n = 0

[Reference] GS P

2.11. ESC ! n

[Name] Select print mode(s)

[Format] ASCII ESC ! n

Hex 1B 21 n

Decimal 27 33 n

[Range] $0 \leq n \leq 255$

[Description] Selects print mode(s) using n as follows:

Table 2-7 Print Modes

Bit	Off/On	Hex	Decimal	Function
0	Off	00	0	Character font A (12 ×24).
	On	01	1	Character font B (9 ×17).
1, 2	-	-	-	Undefined.
3	Off	00	0	Emphasized mode not selected.
	On	08	8	Emphasized mode selected.
4	Off	00	0	Double-height mode not selected.
	On	10	16	Double-height mode selected.
5	Off	00	0	Double-width mode not selected.
	On	20	32	Double-width mode selected.
6	-	-	-	Undefined.
7	Off	00	0	Underline mode not selected.

	On	80	128	Underline mode selected.
--	----	----	-----	--------------------------

[Details]

- When both double-height and double-width modes are selected, quadruple size characters are printed.
- The printer can underline all characters, but cannot underline the space set by HT or 90° clockwise rotated characters.
- The thickness of the underline is that selected by ESC, regardless of the character size.
- When some characters in a line are double or more height, all the characters on the line are aligned at the baseline.
- ESC E can also turn on or off emphasized mode. However, the setting of the last received command is effective.
- ESC —□ can also turn on or off underline mode. However, the setting of the last received command is effective.
- GS ! can also select character size. However, the setting of the last received command is effective.
- Emphasized mode is effective for alphanumeric and Kanji. All print modes except emphasized mode is effective only for alphanumeric.

[Default] n = 0

[Reference] ESC -, ESC E, GS !

2.12. ESC \$ nL nH

[Name] Set absolute print position

[Format]	ASCII	ESC	\$	nL	nH
Hex		1B	24	nL	nH
Decimal		27	36	nL	nH

[Range] $0 \leq nL \leq 255$

$0 \leq nH \leq 255$

[Description] Sets the distance from the beginning of the line to the position at which subsequent characters are to be printed.

The distance from the beginning of the line to the print position is $[(nL + nH \times 256) \times (\text{vertical or horizontal motion unit})]$ inches.

[Details]

- Settings outside the specified printable area are ignored.
- The horizontal and vertical motion units are specified by GS P.
- The GS P command can change the horizontal (and vertical) motion unit. However, the value cannot be less than the minimum horizontal movement amount, and it must be in even units of the minimum horizontal movement amount.
- In standard mode, the horizontal motion unit (x) is used.
- In page mode, horizontal or vertical motion unit differs depending on the starting position of the printable area as follows:
- When the starting position is set to the upper left or lower right of the printable area using ESC T, the horizontal motion unit (x) is used.
- When the starting position is set to the upper right or lower left of the printable area using ESC T, the vertical motion unit (y) is used.

[Reference] ESC \, GS \$, GS \, GS P

2.13. ESC % n

[Name] Select/cancel user-defined character set

[Format]	ASCII	ESC	%	n
----------	-------	-----	---	---

Hex	1B	25	n
Decimal	27	37	n

[Range] $0 \leq nL \leq 255$

[Description]

- Selects or cancels the user-defined character set.
- When the LSB of n is 0, the user-defined character set is canceled.
- When the LSB of n is 1, the user-defined character set is selected.

[Details]

- When the user-defined character set is canceled, the internal character set is automatically selected.
- n is available only for the least significant bit.

[Default] n = 0.

[Reference] ESC &, ESC?

2.14. ESC & y c1 c2 [x1 d1...d(y×x1)]...[xk d1...d(y ×xk)]

[Name] Define user-defined characters

[Format] ASCII ESC & y c1 c2 [x1 d1 ... d (y×x1)] ... [xk d1 ... d (y×xk)]

Hex 1B 26 y c1 c2 [x1 d1 ... d (y×x1)] ... [xk d1 ... d (y ×xk)]

Decimal 27 38 y c1 c2 [x1 d1 ... d (y×x1)] ... [xk d1 ... d (y ×xk)]

[Range] y = 3

$32 \leq c1 \leq c2 \leq 127$

$0 \leq x \leq 12$ standard ASCII font (12×24)

$0 \leq x \leq 9$ compressed ASCII font (9×17) $0 \leq d1 \dots d (y \times xk) \leq 255$

[Description]

- Defines user-defined characters.
- y specifies the number of bytes in the vertical direction.
- c1 specifies the beginning character code for the definition, and c2 specifies the final code.
- x specifies the number of dots in the horizontal direction.

[Details]

- The allowable character code range is from ASCII code <20>H to <7E>H (96 characters).
- It is possible to define multiple characters for consecutive character codes. If only one character is desired, use c1 = c2.
- d is the dot data for the characters. The dot pattern is in the horizontal direction from the left side. Any remaining dots on the right side are blank.
- The data to define a user-defined character is (y×x) bytes.
- Set a corresponding bit to 1 to print a dot or 0 to not print a dot.
- This command can define different user-defined character patterns by each fonts. To select a font, use ESC !
- A user-defined character and a downloaded bit image cannot be defined simultaneously. When this command is executed, the downloaded bit image is cleared.
- The user-defined character definition is cleared when:
 - ① ESC @ is executed.
 - ② ESC ? is executed.
 - ③ FS q is executed.
 - ④ GS * is executed.
 - ⑤ The printer is reset or the power is turned off.
- When the user-defined characters are defined in font B (9×17), only the most significant bit of the 3rd byte of data in vertical direction is effective.

[Default] The internal character set

[Reference] ESC %, ESC ?

[Example]

When font A (12×24) is selected.

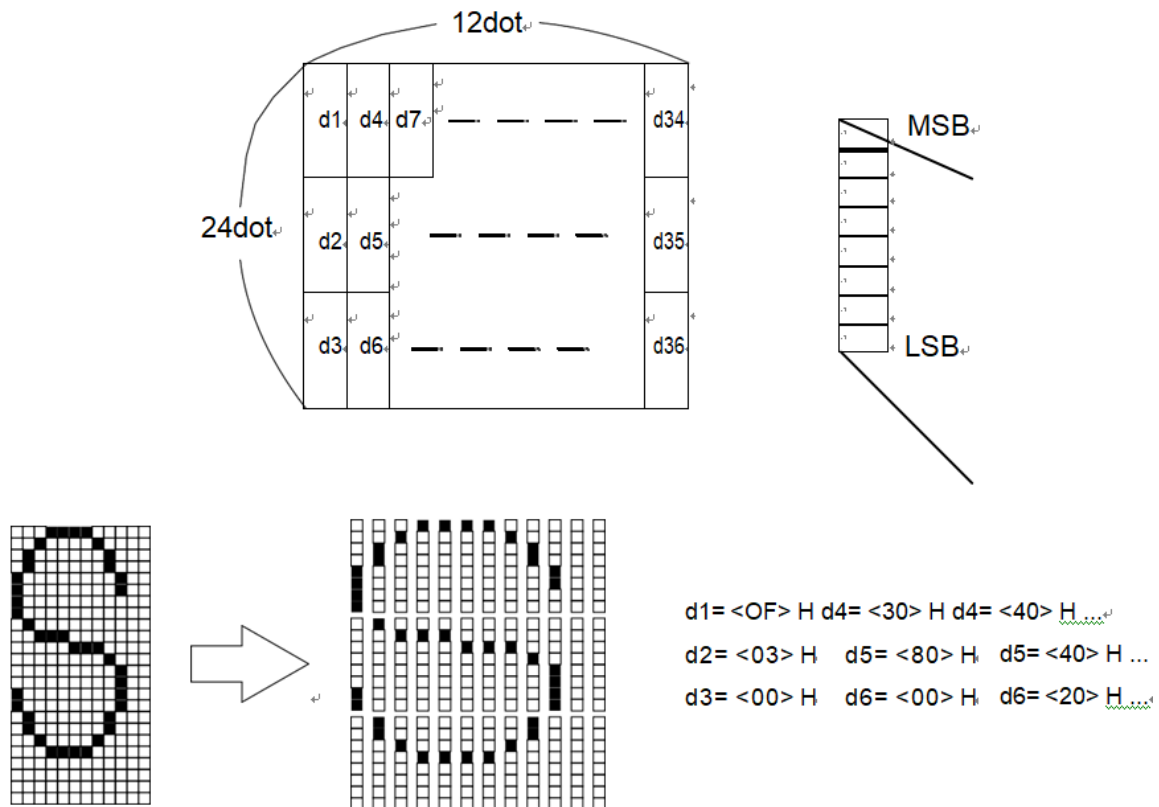


Figure 2-1 Font A (12×24)

- When font B (9×17) is selected.

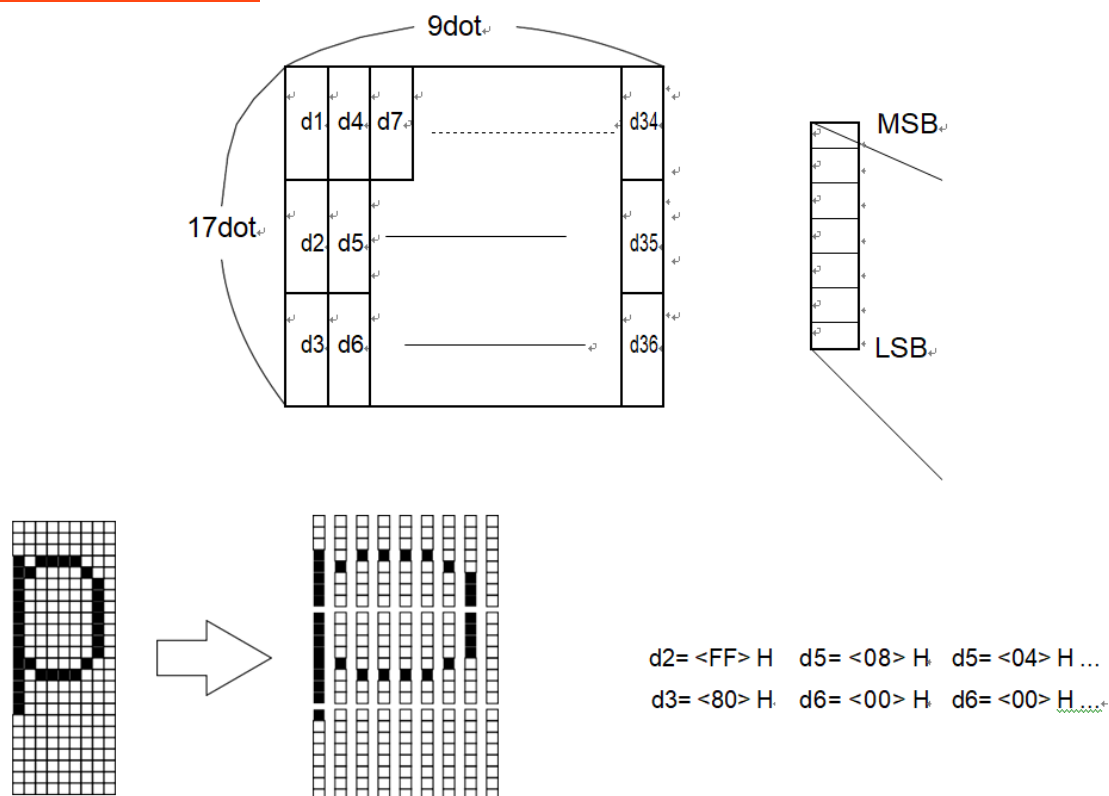


Figure 2-2 Font B (9×17)

2.15. ESC * m nL nH d1... dk

[Name] Select bit-image mode

[Format] ASCII ESC * m nL nH d1...dk

Hex 1B 2A m nL nH d1...dk

Decimal 27 42 m nL nH d1...dk

[Range] m = 0, 1, 32, 33

$0 \leq nL \leq 255$

$0 \leq nH \leq 3$

$0 \leq d \leq 255$

[Description] Selects a bit-image mode using m for the number of dots specified by nL and nH, as follows:

Table 2-8 m

m	Mode	Vertical Direction		Horizontal Direction	
		Number of Dots	Dot Density	Dot Density	Number of Data (K)
0	8-dot single-density	8	67 DPI	100 DPI	$nL + nH \times 256$
1	8-dot double-density	8	67 DPI	200 DPI	$nL + nH \times 256$
32	24-dot single-density	24	200DPI	100DPI	$(nL + nH \times 256) \times 3$
33	24-dot double-density	24	200 DPI	200 DPI	$(nL + nH \times 256) \times 3$

[Details]

- If the values of m is out of the specified range, nL and data following are processed as normal data.
- The nL and nH indicate the number of dots of the bit image in the horizontal direction. The number of dots is calculated by $nL + nH \times 256$.
- If the bit-image data input exceeds the number of dots to be printed on a line, the excess data is ignored.
- d indicates the bit-image data. Set a corresponding bit to 1 to print a dot or to 0 to not print a dot.
- If the width of the printing area set by GS L and GS W less than the width required by the data sent with the ESC * command, the following will be performed on the line in question (but the printing cannot exceed the maximum printable area):
 - ① The width of the printing area is extended to the right to accommodate the amount of data.
 - ② If step ① does not provide sufficient width for the data, the left margin is reduced to accommodate the data.
- After printing a bit image, the printer returns to normal data processing mode.
- This command is not affected by print modes (emphasized, double-strike, underline, character size or white/black reverse printing), except upside-down printing mode.
- The relationship between the image data and the dots to be printed is as follows:
- When 8-dot bit image is selected:

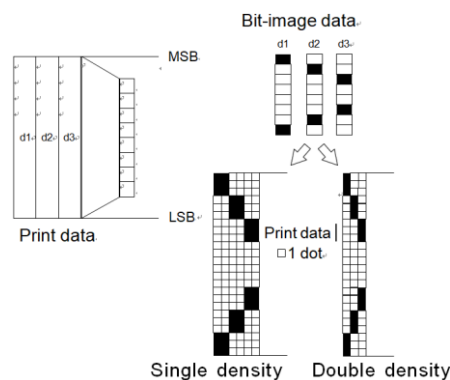


Figure 2-3 8-dot bit image

- When 24-dot bit image is selected:

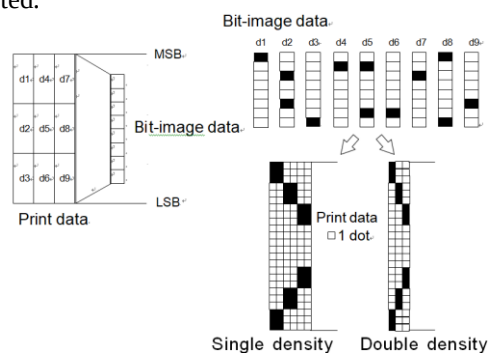


Figure 2-4 24-dot bit image

2.16. ESC – n

[Name] Turn underline mode on/off

[Format] ASCII ESC - n

Hex 1B 2D n

Decimal 27 45 n

[Range] $0 \leq n \leq 2$

$48 \leq n \leq 50$

[Description] Turns underline mode on or off, based on the following values of n:

Table 2-9 n

n	Function
0,48	Turns off underline mode
1,49	Turns on underline mode (1-dot thick)
2,50	Turns on underline mode (2-dots thick)

[Details]

- The printer can underline all characters (including right-side character spacing), but cannot underline the space set by HT.
- The printer cannot underline 90 clockwise rotated characters and white/black inverted characters.
- When underline mode is turned off by setting the value of n to 0 or 48, the following data is not underlined, and the underline thickness set before the mode is turned off does not change. The default underline thickness is 1 dot.
- Changing the character size does not affect the current underline thickness.
- Underline mode can also be turned on or off by using ESC !. Note, however, that the last received command is effective.
- This command does not affect the setting of Kanji characters.

[Default] n = 0

[Reference] ESC !

2.17. ESC 2

[Name] Select default line spacing

[Format] ASCII ESC 2

Hex 1B 32

Decimal 27 50

[Description] Selects 1/6-inch line (approximately 4.23mm) spacing.

[Details] The line spacing can be set independently in standard mode and in page mode.

[Reference] ESC 3

2.18. ESC 3 n

[Name] Set line spacing

[Format] ASCII ESC 3 n

Hex 1B 33 n

Decimal 27 51 n

[Range] $0 \leq n \leq 255$

[Description] Sets the line spacing to [n × vertical or horizontal motion unit] inches.

[Details]

- The line spacing can be set independently in standard mode and in page mode.
- The horizontal and vertical motion units are specified by GS P. Changing the horizontal or vertical motion unit does not affect the current line spacing.
- The GS P command can change the horizontal (and vertical) motion unit. However, the value cannot be less than the minimum vertical movement amount, and it must be in even units of the minimum vertical movement amount.
- In standard mode, the vertical motion unit (y) is used.
- In page mode, this command functions as follows, depending on the starting position of the printable area:

- ① When the starting position is set to the upper left or lower right of the printable area using ESC T, the vertical motion unit (y) is used.
- ② When the starting position is set to the upper right or lower left of the print able area using ESC T, the horizontal motion unit (x) is used.
- The maximum paper feed amount is 1016 mm (40 inches). Even if a paper feed amount of more than 1016 mm (40 inches) is set, the printer feeds the paper only 1016 mm (40 inches).

[Default] Line spacing equivalent to approximately 4.23mm (1/6 inches).

[Reference] ESC 2, GS P

2.19. ESC = n (*)

[Name] Set peripheral device

[Format] ASCII ESC = n
 Hex 1B 3D n
 Decimal 27 61 n

[Range] $1 \leq n \leq 255$

[Description] Selects device to which host computer sends data, using n as follows:

Table 2-10 n

Bit	Off/On	Hex	Decimal	Function
0	Off	00	0	Printer disabled
	On	01	1	Printer enabled
1-7				Undefined

[Details] When the printer is disabled, it ignores all data except for error recovery commands (DLE EOT, DLE ENQ, DLE DC4) until it is enabled by this command.

[Default] n = 1

2.20. ESC ? n

[Name] Cancel user-defined characters

[Range] $32 \leq n \leq 126$

[Description] Cancels user-defined characters.

[Details]

- This command cancels the pattern defined for the character code specified by n. After the user-defined characters is canceled, the corresponding pattern for the internal character is printed.
- This command deletes the pattern defined for the specified code in the font selected by ESC !.
- If a user-defined character has not been defined for the specified character code, the printer ignores this command.

[Reference] ESC &, ESC %

2.21. ESC @

[Name] Initialize printer

[Format] ASCII ESC @
 Hex 1B 40
 Decimal 27 64

[Description] Clears the data in the print buffer and resets the printer mode to the mode that was in effect when the power was turned on.

[Details]

- The DIP switch settings are not checked again.
- The data in the receive buffer is not cleared.
- The macro definition is not cleared.
- The NV bit image data is not cleared.
- The data of the user NV memory is not cleared.

2.22. ESC D n1...nk NUL

[Name] Set horizontal tab positions

[Format]	ASCII	ESC	D	n1...nk	NUL
	Hex	1B	44	n1...nk	00
	Decimal	27	68	n1...nk	0

[Range] $1 \leq n \leq 255$

$0 \leq k \leq 32$

[Description]

- Sets horizontal tab positions.
- n specifies the column number for setting a horizontal tab position from the beginning of the line.
- k indicates the total number of horizontal tab positions to be set.

[Details]

- The horizontal tab position is stored as a value of [character width × n] measured from the beginning of the line. The character width includes the right-side character spacing, and double-width characters are set with twice the width of normal characters.
- This command cancels the previous horizontal tab settings.
- When setting n = 8, the print position is moved to column 9 by sending HT.
- Up to 32 tab positions (k = 32) can be set. Data exceeding 32 tab positions is processed as normal data.
- Transmit [n] k in ascending order and place a NUL code 0 at the end.
- When [n] k is less than or equal to the preceding value [n] k-1, tab setting is finished and the following data is processed as normal data.
- ESC D NUL cancels all horizontal tab positions.
- The previously specified horizontal tab positions do not change, even if the character width changes.
- The character width is memorized for each standard and page mode.

[Default] The default tab positions are at intervals of 8 characters (columns 9, 17, 25,...) for font A (12 × 24).

[Reference] HT

2.23. ESC E n

[Name] Turn emphasized mode on/off

[Format]	ASCII	ESC	E	n
	Hex	1B	45	n
	Decimal	27	69	n

[Range] $0 < n \leq 255$

[Description]

- Turns emphasized mode on or off
- When the LSB of n is 0, emphasized mode is turned off.
- When the LSB of n is 1, emphasized mode is turned on.

[Details]

- Only the least significant bit of n is enabled.
- This command and ESC ! turn on and off emphasized mode in the same way. Be careful when this command is used with ESC !.

[Default] n = 0

[Reference] ESC !

2.24. ESC G n

[Name] Turn on/off double strike mode

[Format]	ASCII	ESC	G	n
	Hex	1B	47	n
	Decimal	27	71	n

[Range] $0 \leq n \leq 255$

[Description]

- Turns double strike mode on or off.
- When the LSB of n is 0, double-strike mode is turned off.
- When the LSB of n is 1, double-strike mode is turned on.

[Details]

- Only the lowest bit of n is enabled.
- Printer output is the same in double-strike mode and in emphasized mode.

[Default] n = 0

[Reference] ESC E

2.25. ESC J n

[Name] Print and feed paper

[Format]	ASCII	ESC	J	n
	Hex	1B	4A	n
	Decimal	27	74	n

[Range] $0 \leq n \leq 255$

[Description] Prints the data in the print buffer and feeds the paper [n ×vertical or horizontal motion unit] inches.

[Details]

- After printing is completed, this command sets the print starting position to the beginning of the line.
- The paper feed amount set by this command does not affect the values set by ESC 2 or ESC 3.
- The horizontal and vertical motion unit are specified by GS P.
- The GS P command can change the vertical (and horizontal) motion unit. However, the value cannot be less than the minimum vertical movement amount, and it must be in even units of the minimum vertical movement amount.
- In standard mode, the printer uses the vertical motion unit (y).
- In page mode, this command functions as follows, depending on the starting position of the printable area:
 - ① When the starting position is set to the upper left or lower right of the printable area using ESC T, the vertical motion unit (y) is used.
 - ② When the starting position is set to the upper right or lower left of the print able area using ESC T, the horizontal motion unit (x) is used.
- The maximum line spacing is 1016mm (40 inches). When the setting value exceeds the maximum, it is converted to the maximum automatically.

[Reference] GS P

2.26. ESC L (*)

[Name] Select page mode

[Format] ASCII ESC L
 Hex 1B 4C
 Decimal 27 76

[Description] Switches from standard mode to page mode.

[Details]

- This command is enabled only when processed at the beginning of a line in standard mode.
- This command has no effect in page mode.
- After printing by FF is completed or by using ESC S, the printer returns to standard mode.
- This command sets the position where data is buffered to the position specified by ESC T within the printing area defined by ESC W.
- This command switches the settings for the following commands (in which the values can be set independently in standard mode and page mode) to those for page mode:
 - ① Set right-side character spacing: ESC SP, FS S
 - ② Select default line spacing: ESC 2, ESC 3
- Only valve settings is possible for the following commands in page mode; these commands are not executed.
 - ① Turn 90°clockwise rotation mode on/off: ESC V
 - ② Select justification: ESC a
 - ③ Turn upside-down printing mode on/off: ESC {
 - ④ Set left margin: GS L
 - ⑤ Set printable area width: GS W
- The following command is ignored in page mode:
 - ① Execute test print: GS (A
- The following command is not available in page mode:
 - ① Print NV bit image: FS p
 - ② Define NV bit image: FS q

2.27. ESC M n (*)

[Name] Select character font

[Format] ASCII ESC M n
 Hex 1B 4D n
 Decimal 27 77 n

[Range] n = 0, 1, 48, 49

[Description] Selects character fonts.

Table 2-11 n

n	Function
0,48	Character font A (12×24) selected.
1,49	Character font B (9×17) selected.

2.28. ESC R n

[Name] Select an international character set

[Format] ASCII ESC R n

Hex 1B 52 n
 Decimal 27 82 n

[Range] $0 \leq n \leq 15$

[Description] Selects an international character set n from the following table:

Table 2-12 International Character Set n

n	Character
0	U.S.A
1	France
2	Germany
3	U.K
4	Denmark
5	Sweden
6	Italy
7	Spain
8	Japan
9	Norway
10	Denmark
11	Spain
12	Latin
13	Korea
14	Slovenia /Croatia
15	Chinese

The character sets for Slovenia/Croatia and China are supported only in the Simplified Chinese model.

[Default] Simplified Chinese model: n = 15.

Models other than the Simplified Chinese model: n = 0.

2.29. ESC S (*)

[Name] Select standard mode

[Format] ASCII ESC S
 Hex 1B 53
 Decimal 27 83

[Description] Switches from page mode to standard mode.

[Details]

- This command is effective only in page mode.
- Data buffered in page mode are cleared.
- This command sets the print position to the beginning of the line.
- The printing area set by ESC W is initialized.
- This command switches the settings for the following commands (in which the values can be set independently in standard mode and page mode) to those for standard mode:
 - ① Set right-side character spacing: ESC SP, FS S
 - ② Select default line spacing: ESC 2, ESC 3
- The following commands are enabled only to set in standard mode.
 - ① Set printing area in page mode: ESC W
 - ② Select print direction in page mode: ESC T
- The following commands are ignored in standard mode.

- ① Set absolute vertical print position in page mode: GS \$
- ② Set relative vertical print position in page mode: GS \
- Standard mode is selected automatically when power is turned on, the printer is reset, or command ESC @ is used.

[Reference] FF, ESC FF, ESC L

2.30. ESC T n (*)

[Name] Select print direction in page mode

[Format] ASCII ESC T n
 Hex 1B 54 n
 Decimal 27 84 n

[Range] $0 \leq n \leq 3$, $48 \leq n \leq 51$

[Description] Selects the print direction and starting position in page mode. n specifies the print direction and starting position as follows:

Table 2-13 n for Page Mode

n	Print Direction	Starting Position
0, 48	Left to right	Upper left
1, 49	Bottom to top	Lower left
2, 50	Right to left	Lower right
3, 51	Top to bottom	Upper right

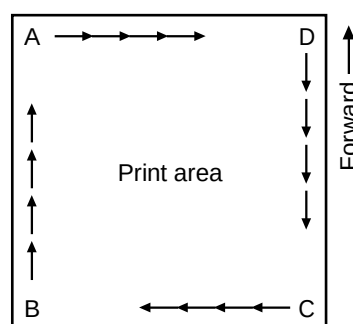


Figure 2-5 Print Direction

[Details]

- When the command is input in standard mode, the printer executes only internal flag operation. This command does not affect printing in standard mode.
- This command sets the position where data is buffered within the printing area set by ESC W.
- Parameters for horizontal or vertical motion units (x or y) differ as follows, depending on the starting position of the printing area:
 - ① If the starting position is the upper left or lower right of the printing area, data is buffered in the direction perpendicular to the paper feed direction: Commands using horizontal motion units: ESC SP, ESC \$, ESC \ Commands using vertical motion units: ESC 3, ESC J, GS \$, GS \
 - ② If the starting position is the upper right or lower left of the printing area, data is buffered in the paper feed direction: Commands using horizontal motion units: ESC 3, ESC J, GS \$, GS \
- Commands using vertical motion units: ESC SP, ESC \$, ESC \

[Default] n = 0

[Reference] ESC \$, ESC L, ESC W, ESC \, GS \$, GS P, GS \

2.31. ESC V n (*)

[Name] Turn 90° clockwise rotation mode on/off

[Format] ASCII ESC V n
 Hex 1B 56 n
 Decimal 27 86 n

[Range] $0 \leq n \leq 1$, $48 \leq n \leq 49$

[Description] Turns 90° clockwise rotation mode on/off n is used as follows:

Table 2-14 Rotation

n	Function
0,48	Turns off 90° clockwise rotation mode
1,49	Turns on 90° clockwise rotation mode

[Details]

- This command affects printing in standard mode. However, the setting is always effective.
- When underline mode is turned on, the printer does not underline 90° clockwise-rotated.
- Double-width and double-height commands in 90° rotation mode enlarge characters in the opposite directions from double height and double width commands in normal mode.

[Default] n = 0

[Reference] ESC !, ESC -

2.32. ESC W xL xH yL yH dxL dxH dyL dyH(*)

[Name] Set printing area in page mode

[Format]	ASCII	ESC	W	xL	xH	yL	yH	dxL	dxH	dyL	dyH
	Hex	1B	57	xL	xH	yL	yH	dxL	dxH	dyL	dyH
	Decimal	27	87	xL	xH	yL	yH	dxL	dxH	dyL	dyH

[Range] $0 \leq xL, xH, yL, yH, dxL, dxH, dyL, dyH \leq 255$ (except $dxL = dxH = 0$ or $dyL = dyH = 0$)

[Description]

The horizontal starting position, vertical starting position, printing area width, and printing area height are defined as x0, y0, dx (inch), dy (inch), respectively. Each setting for the printing area is calculated as follows:

$$x0 = [(xL + xH \times 256) \times (\text{horizontal motion unit})]$$

$$y0 = [(yL + yH \times 256) \times (\text{vertical motion unit})]$$

$$dx = [(dxL + dxH \times 256) \times (\text{horizontal motion unit})]$$

$$dy = [(dyL + dyH \times 256) \times (\text{vertical motion unit})]$$

The printing area is set as shown in the figure below.

[Details]

- If this command is input in standard mode, the printer executes only internal flag operation. This command does not affect printing in standard mode.
- If the horizontal or vertical starting position is set outside the printable area, the printer stops command processing and processes the following data as normal data.
- If the printing area width or height is set to 0, the printer stops command processing and processes the following data as normal data.
- This command sets the position where data is buffered to the position specified by ESC T within the printing area.
- If (horizontal starting position + printing area width) exceeds the printable area, the printing area width is automatically set to (horizontal printable area - horizontal starting position).
- If (vertical starting position + printing area height) exceeds the printable area, the printing area height is automatically set to (vertical printable area - vertical starting position).
- The horizontal and vertical motion unit are specified by **GS P**. Changing the horizontal or vertical motion unit does not affect the current printing area.
- The **GS P** command can change the horizontal (and vertical) motion unit.
However, the value cannot be less than the minimum horizontal movement amount, and it must be in even units of minimum horizontal movement amount.

- Use the horizontal motion unit (x) for setting the horizontal starting position and printing area width, and use the vertical motion unit (y) for setting the vertical starting position and printing area height.
- When the horizontal starting position, vertical starting position, printing area width, and printing area height are defined as X, Y, Dx, and Dy respectively, the printing area is set as shown in the figure below.

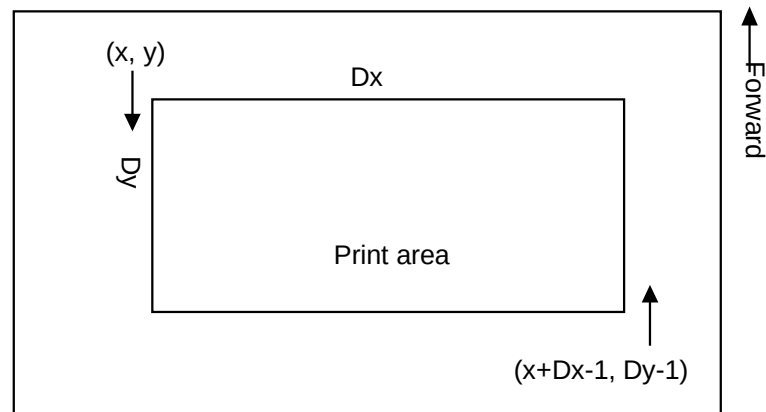


Figure 2-6 X, Y setting of Print Area

- This printable area for this printer is approximately 72 mm in the horizontal direction and approximately 117.3 mm (1662/360 inches) in the vertical direction.

[Default] xL = xH = yL = yH = 0

dxL = 0, dxH = 2, dyL = 126, dyH = 6

[Reference] CAN, ESC L, ESC T, GS P

2.33. ESC \ nL nH

[Name] Set relative print position

[Format]	ASCII	ESC	\	nL	nH
Hex		1B	5C	nL	nH
Decimal		27	92	nL	nH

[Range] $0 \leq nL \leq 255$, $0 \leq nH \leq 255$

[Description]

- Sets the print starting position based on the current position by using the horizontal or vertical motion unit.
- This command sets the distance from the current position to $[(nL + nH \times 256) \times \text{horizontal or vertical motion unit}]$

[Details]

- Any setting that exceeds the printable area is ignored.
- When pitch N is specified to the right: $nL + nH \times 256 = N$
- When pitch N is specified to the left (the negative direction), use the complement of 65536. When pitch N is specified to the left: $nL + nH \times 256 = 65536 - N$
- The print starting position moves from the current position to $[N \times \text{horizontal or vertical motion unit}]$
- The horizontal and vertical motion unit are specified by GS P.
- The GS P command can change the horizontal (and vertical) motion unit. However, the value cannot be less than the minimum horizontal movement amount, and it must be in even units of the minimum horizontal movement amount.
- In standard mode, the horizontal motion unit is used.
- In page mode, the horizontal or vertical motion unit differs as follows, depending on the starting point of the printing area:

- ①When the starting position is set to the upper left or lower right of the printable area using ESC T, the horizontal motion unit (x) is used.
- ②When the starting position is set to the upper right or lower left of the printable area using ESC T, the vertical motion unit (y) is used.

[Reference] ESC \$, GS P

2.34. ESC a n

[Name] Select justification

[Format] ASCII ESC a n
 Hex 1B 61 n
 Decimal 27 97 n

[Range] $0 \leq n \leq 2$, $48 \leq n \leq 50$

[Description] Aligns all the data in one line to the specified position n selects the justification as follows:

Table 2-15 Justification n

n	Justification
0,48	Left justification
1,49	Centering
2,50	Right justification

[Details]

- The command is enabled only when processed at the beginning of the line in standard mode.
- If this command is input in page mode, the printer performs only internal flag operations.
- This command has no effect in page mode.
- This command executes justification in the printing area.
- This command justifies the space area according to HT, ESC \$ or ESC \.

[Default] n = 0

[Example]

Left justification	Centering	Right justification
ABC ABCD ABCDE	ABC ABCD ABCDE	ABC ABCD ABCDE

Figure 2-7 Justification

2.35. ESC c 3 n (*)

[Name] Select paper sensor(s) to output paper end signals

[Format] ASCII ESC c 3 n
 Hex 1B 63 33 n
 Decimal 27 99 51 n

[Range] $0 \leq n \leq 255$

[Description]

Selects the paper sensor(s) to output paper end signals. Each bit of n is used as follows:

Table 2-16 The bit map of Paper Sensors

Bit	Off/On	Hex	Decimal	Function
0	Off	00	0	Paper roll near-end sensor disabled
	On	01	1	Paper roll near-end sensor enabled
1	Off	00	0	Paper roll end sensor disabled
	On	02	2	Paper roll near-end sensor enabled
2	Off	00	0	Paper roll end sensor disabled
	On	04	4	Paper roll near-end sensor enabled
3	Off	00	0	Paper roll end sensor disabled
	On	08	8	Paper roll near-end sensor enabled
4 - 7	-	-	-	Undefined

[Details]

- It is possible to select multiple sensors to output signals. Then, if any of the sensors detects a paper end, the paper end signal is output.
- The command is available only with a parallel interface and is ignored with a serial interface.
- Sensor is switched when executing this command. The paper end signal switching be delayed depending on the receive buffer state.
- If either bit 0 or bit 1 is on, the paper roll near-end sensor is selected as the paper sensor outputting paper-end signals.
- If either bit 2 or bit 3 is on, the paper roll end sensor is selected as the paper sensor outputting paper-end signals.
- When all the sensors are disabled, the paper end signal always outputs a paper present status.

[Default] n = 15

2.36. ESC c 4 n (*)

[Name] Select paper sensor(s) to stop printing

[Format] ASCII ESC c 4 n

Hex 1B 63 34 n

Decimal 27 99 52 n

[Range] $0 \leq n \leq 255$

[Description] Selects the paper sensor(s) used to stop printing when a paper-end is detected, using n as follows:

Table 2-17 The bit map of Paper Sensors

Bit	Off/On	Hex	Decimal	Function
0	Off	00	0	Paper roll near-end sensor disabled
	On	01	1	Paper roll near-end sensor enabled
1	Off	00	0	Paper roll end sensor disabled

	On	02	2	Paper roll near-end sensor enabled
2 - 7	-	-	-	Undefined

[Details]

- When a paper sensor is enabled with this command, printing is stopped only when the corresponding paper is selected for printing.
- When a paper-end is detected by the paper roll sensor, the printer goes offline after printing stops.
- When either bit 0 or 1 is on, the printer selects the paper roll near-end sensor for the paper sensor to stop printing.

[Default] n = 0

2.37. ESC C 5 n

[Name] Enable/disable panel buttons

[Format] ASCII ESC c 5 n
 Hex 1B 63 35 n
 Decimal 27 99 53 n

[Range] $0 \leq n \leq 255$

[Description] Enables or disables the panel buttons.

- When the LSB of n is 0, the panel buttons are enabled.
- When the LSB of n is 1, the panel buttons are disabled.

[Details]

Only the lowest bit of n is valid.

- When the panel buttons are disabled, none of them are usable when the printer cover is closed.
- In this printer, the panel buttons are the FEED button.
- In the macro ready mode, the FEED button is enabled regardless of the settings of this command; however, the paper cannot be fed by using these buttons.

[Default] n = 0

2.38. ESC d n

[Name] Print and feed n lines

[Format] ASCII ESC d n
 Hex 1B 64 n
 Decimal 27 100 n

[Range] $0 \leq n \leq 255$

[Description] Prints the data in the print buffer and feeds n lines.

[Details]

- This command sets the print starting position to the beginning of the line.
- This command does not affect the line spacing set by ESC 2 or ESC 3.
- The maximum paper feed amount is 1016 mm (40 inches). If the paper feed amount (nx line spacing) of more than 1016 mm (40 inches) is specified, the printer feeds the paper only 1016 mm (40 inches).

[Reference] ESC 2, ESC 3

2.39. ESC p m t1 t2

[Name] Generate pulse

[Format] ASCII ESC p m t1 t2
 Hex 1B 70 m t1 t2
 Decimal 27 112 m t1 t2

[Range] m = 0, 1, 48, 49

$0 \leq t1 \leq 255, 0 \leq t2 \leq 255$

[Description] Outputs the pulse specified by t1 and t2 to connector pin m as follows:

Table 2-18 m for connector pin

m	Connector pin
0,48	Drawer kick-out connector pin 2.
1,49	Drawer kick-out connector pin 5.

[Details]

- The pulse ON time is [t1×2 ms] and the OFF time is [t2×2 ms].
- If t2 < t1, the OFF time is [t1x 2 ms] [Reference] DLE Dc4

2.40. ESC t n

[Name] Select character code table

[Format] ASCII ESC t n
 Hex 1B 74 n
 Decimal 27 116 n

[Range] $0 \leq n \leq 10, 16 \leq n \leq 21$

[Description] Selects a page n from the character code table.

Table 2-19 Character Code

n	Page
0	PC437 [U.S.A. Standard Europe]
1	Katakana
2	PC850: Multilingual
3	PC860: Portuguese
4	PC863 [Canadian French]
5	PC865:Nodic
6	West Europe
7	Greek
8	Hebrew

9	PC755:East Europe
10	Iran
16	WPC1252
17	PC866:Cyrillic#2
18	PC852:Latin2
19	PC858
20	Iran II
21	Latvian

[Default] n = 0

2.41. ESC { n

[Name] Turns on/off upside-down printing mode

[Format] ASCII ESC { n

Hex 1B 7B n

Decimal 27 123 n

[Range] $0 \leq n \leq 255$

[Description] Turns upside down printing mode on or off.

- When the LSB of n is 0, upside-down printing mode is turned off.
- When the LSB of n is 1, upside-down printing mode is turned on.

[Details]

- Only the lowest bit of n is valid.
- This command is enabled only when processed at the beginning of a line in standard mode.
- When this command is input in page mode, the printer performs only internal flag operations.
- This command does not affect printing in page mode.
- In upside-down printing mode, the printer rotates the line to be printed by 180° and then prints it.

[Default] n = 0

[Example]

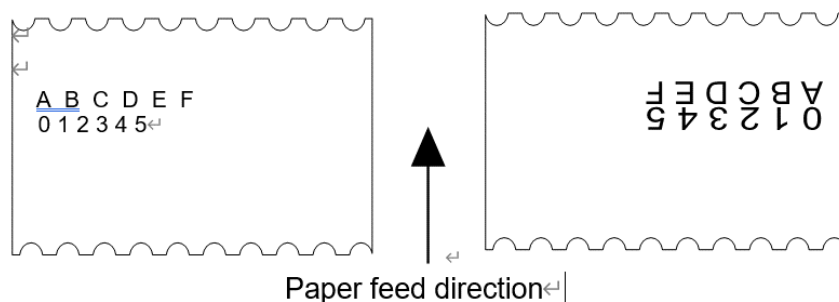


Figure 2-8 Paper Feed Direction

2.42. FS p n m (*)

[Name] Print NV bit image

[Format] ASCII FS p n m

Hex 1C 70 n m

Decimal 28 112 n m

[Range] $0 \leq n \leq 255$

$0 \leq m \leq 3, 48 \leq m \leq 51$

[Description]

- Prints a NV bit image n using the mode specified by m.

Table 2-20 m for NV Bit Image

m	Mode	Vertical Dot Density	Horizontal Dot Density
0, 48	Normal	200 dpi	200 dpi
1, 49	Double-width	200 dpi	100 dpi
2, 50	Double-height	100 dpi	200 dpi
3, 51	Quadruple	100 dpi	100 dpi

[dpi: dots per 25.4 mm {1"}]

- n is the number of the NV bit image (defined using the FS q command).
- m specifies the bit image mode.

[Details]

- NV bit image means a bit image which is defined in a non-volatile memory by **FS q** and printed by **FS p**.
- This command is not effective when the specified NV bit image has not been defined.
- In standard mode, this command is effective only when there is no data in the print buffer.
- In page mode, this command is not effective.
- This command is not affected by print modes (emphasized, double-strike, underline, character size, white/black reverse printing, or 90° rotated characters, etc.), except upside-down printing mode.
- If the printing area width set by **GS L** and **GS W** for the NV bit image is less than one vertical line, the following processing is performed only on the line in question. However, in NV bit image mode, one vertical line means 1 dot in normal mode (m=0, 48) and in double-height mode (m=2, 50), and it means 2 dots in double-width mode (m=1, 49) and in quadruple mode (m=3, 51).
 - ①The printing area width is extended to the right in **NV** bit image mode up to one line vertically. In this case, printing does not exceed the printable area.
 - ②If the printing area width cannot be extended by one line vertically, the left margin is reduced to accommodate one line vertically.
- If the downloaded bit-image to be printed exceeds one line, the excess data is not printed.
- This command feeds dots (for the height n of the **NV** bit-image) in normal and double-width modes, and (for the height n 2 of the NV bit-image) in double-height and quadruple modes, regardless of the line spacing specified by **ESC 2** or **ESC 3**.
- After printing the bit image, this command sets the print position to the beginning of the line and processes the data that follows as normal data.

[References] **ESC**, **FS q**, **GS /**, **GS v 0**

2.43. FS q n [xL xH yL yH d1...dk]1...[xL xH yL yH d1...dk]n (*)

[Name] Define NV bit image

[Format] ASCII FS q n [xL xH yL yH d1...dk]1...[xL xH yL yH d1...dk]n

Hex 1C 71 n [xL xH yL yH d1...dk]1...[xL xH yL yH d1...dk]n

Decimal 28 113 n [xL xH yL yH d1...dk]1...[xL xH yL yH d1...dk]n

[Range] $0 \leq n \leq 255$

$0 \leq xL \leq 255$

$0 \leq xH \leq 3$ (when $1 \leq (xL + xH \times 256) \leq 1023$)

$$0 \leq yL \leq 255$$

$$0 \leq yL \leq 1 \text{ (when } 1 \leq (yL + yH \times 256) \leq 288 \text{)}$$

$$0 \leq d \leq 255$$

$$k = (xL + xH \times 256) \times (yL + yH \times 256) \times 8$$

Total defined data area = 0.5M bits (64K bytes)

[Description] Define the NV bit image specified by n.

- n specifies the number of the defined NV bit image.
- xL, xH specifies $(xL + xH \times 256) \times 8$ dots in the horizontal direction for the NV bit image you are defining.
- yL, yH specifies $(yL + yH \times 256) \times 8$ dots in the vertical direction for the NV bit image you are defining.

[Details]

- This command cancels all NV bit images that have already been defined by this command. The printer cannot redefine only one of several data definitions previously defined. In this case, all data needs to be sent again.
- From the beginning of the processing of this command till the finish of hardware reset, mechanical operations (including initializing the position of the printer head when the cover is open, paper feeding by using the FEED button, etc.) cannot be performed.
- During processing this command, the printer is in BUSY when writing the data to the NV user memory and stops receiving data. So it is prohibited to transmit the data including the real-time commands during the execution of this command.
- NV bit image means a bit image which is defined in a non-volatile memory by FS q and printed by FS p.
- In standard mode, this command is effective only when processed at the beginning of the line.
- In page mode, this command is not effective
- This command is effective when 7 bytes <FS<yH> is processed as a normal value.
- When the amount of data exceeds the capacity left in the range defined by xL, xH, yL, yH, the printer processes xL, xH, yL, yH out of the defined range.
- In the first group of NV bit images, when any of the parameters xL, xH, yL, yH is out of the definition range, this command is disabled.
- In groups of NV bit images other than the first one, when the printer processes xL, xH, yL, yH out of the defined range, it stops processing this command and starts writing into the NV images. At this time, NV bit images that haven't been defined are disabled (undefined), but any NV bit images before that are enabled.
- The d indicates the definition data. In data (d) a 1 bit specifies a dot to be printed and a 0 bit specifies a dot not to be printed.
- This command defines n as the number of a NV bit image. Numbers rise in order from NV bit image 01H. Therefore, the first data group [xL xH yL yH d1...dk] is NV bit image 01H, and the last data group [xL xH yL yH d1...dk] is NV bit image n. The total agrees with the number of NV bit images specified by command FS p.
- A definition data of a NV bit image consists of [xL xH yL yH d1...dk]. Therefore, when only one NV bit image is defined n=1, the printer processes a data group [xL xH yL yH d1...dk] once. The printer uses ([data: $(xL + xH \times 256) \times (yL + yH \times 256) \times 8$] + [header :4]) bytes of NV memory.
- The definition area in this printer is a maximum of 0.5M bits (64K bytes). This command can define several NV bit images, but cannot define a bit image data whose total capacity [bit image data + header] exceeds 0.5M bits (64K bytes).
- The printer is busy immediately before writing into NV memory.
- The printer does not transmit ASB status and perform status detection during processing of this command even when ASB is specified.
- When this command is received during macro definition, the printer ends macro definition, and begins performing this command.
- Once a NV bit image is defined, it is not erased by performing ESC @, reset, and power off.
- This command performs only definition of a NV bit image and does not perform printing. Printing the NV bit image is performed by the FS p command.
- Frequent write command execution may cause damage the NV memory.
- Therefore, it is recommended to write the NV memory 10 times or less a day.

- The printer performs a hardware reset after the procedure to place the image into the NV memory. Therefore, user defined characters, downloaded bit images, and macros should be defined only after completing this command. The printer clears the receive and print buffers and resets the mode to the mode that was in effect at power on. At this time, DIP switch settings are checked again.

[Reference] **FS p**

[Example] When $xL = 64$, $xH = 0$, $yL = 96$, $yH = 0$

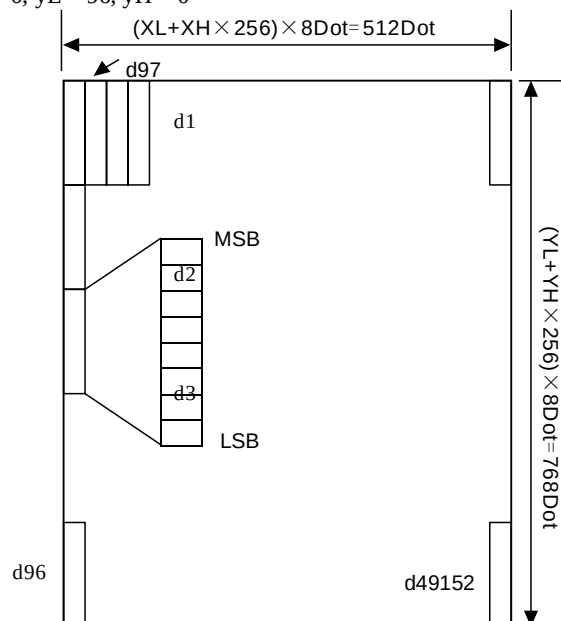


Figure 2-9 NV Bit Image Example

2.44. GS ! n (*)

[Name] Select character size

[Format] ASCII GS ! N

Hex 1D 21 n

Decimal 29 33 n

[Range] $0 \leq n \leq 255$ ($1 \leq \text{vertical number of times} \leq 8$, $1 \leq \text{horizontal number of times} \leq 8$)

[Description] Selects the character height using bits 0 to 2 and selects the character width using bits 4 to 7, as follows:

Table 2-21 Character Size Selection

Bit	Functions
0-3	Character height selection. See Table 2-22.
4-7	Character width selection. See Table 2-23.

Table 2-22 Character Height Selection

Hex	Decimal	width
00	0	1(normal)
10	16	2(doubie-height)
20	32	3
30	48	4
40	64	5

Table 2-23 Character Width Selection

Hex	Decimal	width
00	0	1(normal)
10	16	2(doubie-width)
20	32	3
30	48	4
40	64	5

50	80	6
60	96	7
70	112	8

50	80	6
60	96	7
70	112	8

[Details]

- This command is all characters (alphanumeric and Kanji) effective except for HRI characters.
- If n is outside of the defined range, this command is ignored.
- In standard mode, the vertical direction is the paper feed direction, and the horizontal direction is perpendicular to the paper feed direction. However, when character orientation changes in 90° clockwise-rotation mode, the relationship between vertical and horizontal directions is reversed.
- In page mode, vertical and horizontal directions are based on the character orientation.
- When characters are enlarged with different sizes on one line, all the characters on the line are aligned at the baseline. The **ESC !** command can also turn double-width and double-height modes on or off. However, the setting of the last received command is effective.

[Default] n = 0

[Reference] ESC !

2.45. GS \$ nL nH (*)

[Name] Set absolute vertical print position in page mode

[Format] ASCII GS \$ nL nH
 Hex 1D 24 nL nH
 Decimal 29 36 nL nH

[Range] $0 \leq nL \leq 255, 0 \leq nH \leq 255$

[Description]

- Sets the absolute vertical print starting position for buffer character data in page mode.
- This command sets the absolute print position to $[(nL + nH \times 256) \times (\text{vertical or horizontal motion unit})]$ inches.

[Details]

- This command is effective only in page mode.
- If the $[(nL + nH \times 256) \times (\text{vertical or horizontal motion unit})]$ exceeds the specified printing area, this command is ignored.
- The horizontal starting buffer position does not move.
- The reference starting position is that specified by ESC T.
- This command operates as follows, depending on the starting position of the printing area specified by ESC T: G
 - ①When the starting position is set to the upper left or lower right, this command sets the absolute position in the vertical direction.
 - ②When the starting position is set to the upper right or lower left, this command sets the absolute position in the horizontal direction.
- The horizontal and vertical motion unit are specified by GS P.
- The GS P command can change the horizontal and vertical motion unit.
- However, the value cannot be less than the minimum horizontal movement amount, and it must be in even units of the minimum horizontal movement amount.

[Reference] ESC \$, ESC T, ESC W, ESC \, GS P, GS \

2.46. GS * x y d1...d(x × y × 8)

[Name] Define downloaded bit image

[Format] ASCII GS * xy d1 ... d (x × y × 8)

Hex 1D 2A xy d1 ... d (x × y × 8)

Decimal 29 42 xy d1 ... d (x × y × 8)

[Range] $1 \leq x \leq 255$, $1 \leq y \leq 48$ $x \times y \leq 912$ $0 \leq d \leq 255$

[Description] Defines a downloaded bit image using the number of dots specified by x and y

- x specifies the number of dots in the horizontal direction.
- y specifies the number of dots in the vertical direction.

[Details]

- The number of dots in the horizontal direction is $x \times 8$, in the vertical direction it is $y \times 8$.
- If $x \times y$ is out of the specified range, this command is disabled.
- The d indicates bit-image data. Data (d) specifies a bit printed to 1 and not printed to 0. The downloaded bit image definition is cleared when: ESC @ is executed, or ESC & is executed, or FS q is executed. Printer is reset or the power is turned off.
- The following figure shows the relationship between the downloaded bit image and the printed data.

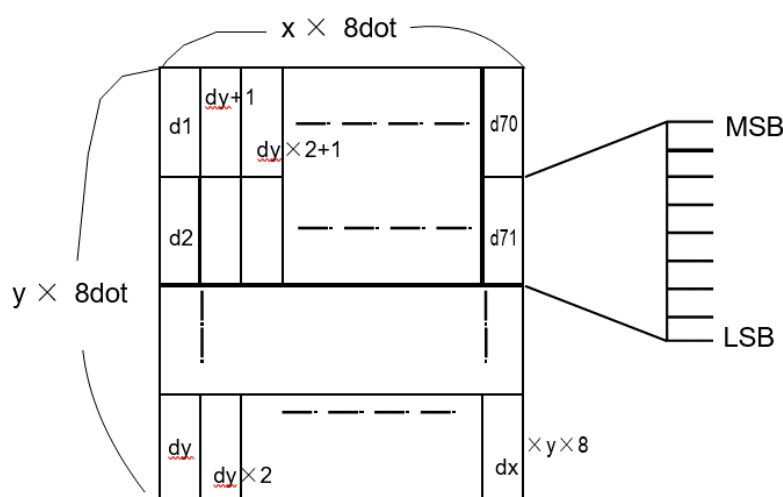


Figure 2-10 The Relationship between download bit image and printed data

[Reference] GS /

2.47. GS / m

[Name] Print downloaded bit image

[Format] ASCII GS / m

Hex 1D 2F m

Decimal 29 47 m

[Range] $0 \leq m \leq 3$, $48 \leq m \leq 51$

[Description] Prints a downloaded bit image using the mode specified by m. m selects a mode from the table below:

Table 2-24 Mode Selection for Density

m	Mode	Vertical Dot Density (DPI)	Horizontal Dot Density (DPI)
0, 48	Normal	200	200
1, 49	Double-width	200	100

2, 50	Double-height	100	200
3, 51	Quadruple	100	100

[Details]

- This command is ignored if a downloaded bit image has not been defined.
- In standard mode, this command is effective only when there is no data in the print buffer.
- This command has no effect in the print modes (emphasized, double-strike, underline, character size, or white/black reverse printing), except for upside down printing mode.
- If the downloaded bit-image to be printed exceeds the printable area, the excess data is not printed.
- Refer to Figure 3.12.3 for the downloaded bit image development position in page mode.
- If the printing area width set by GS L and GS W is less than one line in vertical, the following processing is performed only on the line in question:
- The printing area width is extended to the right up to one line in vertical. In this case, printing does not exceed the printable area.
- If the printing area width cannot be extended by one line in vertical, the left margin is reduced to accommodate one line in vertical.

[Reference] GS *

2.48. GS (A pL pH n m

[Name] Execute print data hex dump

[Format] ASCII GS (A pL pH n m
 Hex 1D 28 41 pL pH n m
 Decimal 29 40 65 pL pH n m

[Range] pL = 2, pH = 0;

n = 0,48; m = 1,49;

2.49. GS: (*)

[Name] Start/end macro definition

[Format] ASCII GS :
 Hex 1D 3A
 Decimal 29 58

[Description] Starts or ends macro definition.

[Details] • Macro definition starts when this command is received during normal operation. Macro definition ends when this command is received during macro definition.

When GS ^ is received during macro definition, the printer ends macro definition and clears the definition.

Macro is not defined when the power is turned on.

The defined contents of the macro are not cleared by ESC @. Therefore, ESC @ can be included in the contents of the macro definition.

If the printer receives GS : again immediately after previously receiving GS : the printer remains in the macro undefined state.

The contents of the macro can be defined up to 2048 bytes. If the macro definition exceeds 2048 bytes, excess data is not stored.

[Reference] GS ^

2.50. GSBn (*)

[Name] Turn white/black reverse printing mode

[Format] ASCII GS B n
 Hex 1D 42 n
 Decimal 29 66 n

[Range] $0 \leq n \leq 255$

[Description] Turns on or off white/black reverse printing mode.

- When the LSB of n is 0, white/black reverse mode is turned off.
- When the LSB of n is 1, white/black reverse mode is turned on.

[Details]

- Only the lowest bit of n is valid.
- This command is available for built-in characters and user-defined characters.
- When white/black reverse printing mode is on, it also applied to character spacing set by ESC SP.
- This command does not affect bit image, user-defined bit image, bar code, HRI characters, and spacing skipped by HT, ESC \$, and ESC \.
- This command does not affect the space between lines.
- White/black reverse mode has a higher priority than underline mode. Even if underline mode is on, it is disabled (but not canceled) when white/black reverse mode is selected.

[Default] n = 0

2.51. GS H n

[Name] Select printing position for HRI characters

[Format] ASCII GS H n
 Hex 1D 48 n
 Decimal 29 72 n

[Range] $0 \leq n \leq 3, 48 \leq n \leq 51$

[Description] Selects the printing position of HRI characters when printing a bar code. n selects the printing position as follows:

Table 2-25 Print Position

n	Print Position
0,48	Do not print
1,49	Top of the bar code
2,50	Below the barcode
3,51	The barcode is printed below

HRI indicates Human Readable Interpretation.

[Details] HRI characters are printed using the font specified by GS f.

[Default] n = 0

[Reference] GS f, GS k

2.52. GS L nL nH

[Name] Set left margin

[Format] ASCII GS L nL nH
 Hex 1D 4C nL nH

Decimal 29 76 nL nH

[Range] $0 \leq nL \leq 255$ $0 \leq nH \leq 255$

[Description]

- Sets the left margin using nL and nH.
- The left margin is set to $[(nL + nH \times 256) \times \text{horizontal motion unit}]$ inches. Printable area Left margin Printing area width

[Details]

- This command is effective only processed at the beginning of the line in standard mode.
- If this command is input in page mode, the printer performs only internal flag operations.
- This command does not affect printing in page mode.
- If the setting exceeds the printable area, the maximum value of the printable area is used.
- The horizontal and vertical motion units are specified by GS P. Changing the horizontal and vertical motion unit does not affect the current left margin.
- The horizontal motion unit (x) is used for calculating the left margin. The calculated result is truncated to the minimum value of the mechanical pitch.

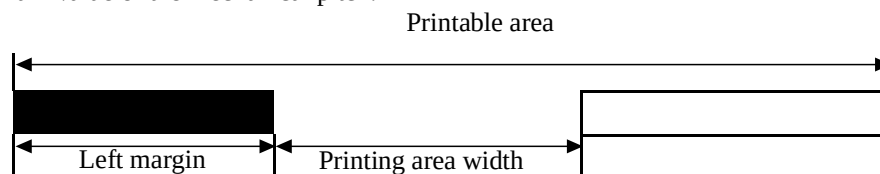


Figure 2-11 Printing Area Width

[Default] nL = 0, nH = 0

[Reference] GS P, GS W

2.53. GS P x y (*)

[Name] Set horizontal and vertical motion units

[Format] ASCII GS P x y

Hex 1D 50 x y

Decimal 29 80 x y

[Range] $0 \leq x \leq 255$

$0 \leq y \leq 255$

[Description] Sets the horizontal and vertical motion units to approximately $25.4/x$ mm $\{1/x$ inches $\}$ and approximately $25.4/y$ mm $\{1/y$ inches $\}$, respectively. When x and y are set to 0, the default setting of each value is used.

[Details]

- The horizontal direction is perpendicular to the paper feed direction and the vertical direction is the paper feed direction.
- In standard mode, the following commands use x or y, regardless of character rotation (upside-down or 90° clockwise rotation): ①Commands using x: ESC SP, ESC \$, ESC \, FS S, GS L, GS W ②Commands using y: ESC 3, ESC J, GS V
- In page mode, the following command use x or y, depending on character orientation:
 - ① When the print starting position is set to the upper left or lower right of the printing area using ESC T (data is buffered in the direction perpendicular to the paper feed direction): Commands using x: ESC SP, ESC \$, ESC W, ESC \, FS S Commands using y: ESC 3, ESC J, ESC W, GS \$, GS \, GS V
 - ② When the print starting position is set to the upper right or lower left of the printing area using ESC T (data is buffered in the paper feed direction): Commands using x: ESC 3, ESC J, ESC W, GS \$, GS \ Commands using y: ESC SP, ESC \$, ESC W, ESC \, FS S, GS V
- The command does not affect the previously specified values.

- The calculated result from combining this command with others is truncated to the minimum value of the mechanical pitch.

[Default] x = 180, y = 360

[Reference] ESC SP, ESC \$, ESC 3, ESC J, ESC W, ESC \, GS \$, GS L, GS V, GS W, GS \

2.54. ① GS V m ② GS V m n

[Name] Select cut mode and cut paper

[Format] ① ASCII GS V m
 Hex 1D 56 m
 Decimal 29 86 m
 ② ASCII GS V m n
 Hex 1D 56 m n
 Decimal 29 86 m n

[Range] ① m=0, 48, m =1,49

② m = 66, $0 \leq n \leq 255$

[Description] Selects a mode for cutting paper and executes paper cutting. The value of m selects the mode as follows:

Table 2-26 Cut Mode

m	Print mode
0,48	one point left uncut
1,49	Partial cut
66	Feeds paper (cutting position + [n × (vertical motion unit)]), and cuts the paper partially (one point left uncut).

[Details for ① and ②]

This command is effective only processed at the beginning of a line.

[Details for ①] Only the partial cut is available; there is no full cut.

[Details for ②]

- When n = 0, the printer feeds the paper to the cutting position and cuts it.
- When n = 0, the printer feeds the paper to (cutting position + [n × vertical motion unit]) and cuts it.
- The horizontal and vertical motion unit are specified by GS P.
- The paper feed amount is calculated using the vertical motion unit (y). However, the value cannot be less than the minimum horizontal movement amount, and it must be in even units of the minimum horizontal movement amount.

2.55. GS W nL nH

[Name] Set printing area width

[Format] ASCII GS W nL nH
 Hex 1D 57 nL nH
 Decimal 29 87 nL nH

[Range] $0 \leq nL \leq 255$

$$0 \leq nH \leq 255$$

[Description] Sets the printing area width to the area specified by nL and nH.

The printing area width is set to $[(nL + nH \times 256) \times \text{horizontal motion unit}]$ inches.

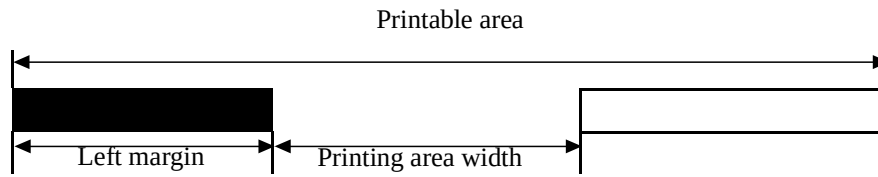


Figure 2-12 Printing Area Width

[Details]

- This command is effective only processed at the beginning of the line.
- In page mode, the printer performs only internal flag operations.
- This command does not affect printing in page mode.
- If the [left margin + printing area width] exceeds the printable area, [printable area width - left margin] is used.
- The horizontal and vertical motion units are specified by GS P. Changing the horizontal and vertical motion units does not affect the current left margin.
- The horizontal motion unit (x) is used for calculating the printing area width. The calculated result is truncated to the minimum value of the mechanical pitch.
- If the width set for the printing area is less than the width of one character, when the character data is developed, the following processing is performed:
 - The printing area width is extended to the right to accommodate one character.
 - If the printing area width cannot be extended sufficiently, the left margin is reduced to accommodate one character. If the printing area width cannot be extended sufficiently, the right space is reduced.
- If the width set for the printing area is less than one line in vertical, the following processing is performed only on the line in question when data other than character data (e.g., bit image, user-defined bit image) is developed:
 - The printing area width is extended to the right to accommodate one line in vertical for the bit image within the printable area.
 - If the printing area width cannot be extended sufficiently, the left margin is reduced to accommodate one line in vertical.
- The commands which set the printing area width for bit image printing and its minimum widths are as follows:
 - Bit image (ESC *):
Single density mode = 2 dots Double density mode = 1 dot
 - Downloaded bit image (GS /):
Double width mode or Quadruple mode = 2 dots Normal mode or Double-height mode = 1 dot
 - NV bit image (FS p):
Double width mode or Quadruple mode = 2 dots Normal mode or Double-height mode = 1 dot
 - Raster bit image (GS r 0):
Double width mode or Quadruple mode = 2 dots Normal mode or Double-height mode = 1 dot

[Default] nL = 0, nH = 2. For 58mm paper width model; nL = 104, nH = 1

[Reference] GS L, GS P

2.56. GS \ nL nH

[Name] Set relative vertical print position in page mode

[Format] ASCII	GS	\	nL	nH
Hex	1D	5C	nL	nH
Decimal	29	92	nL	nH

[Range] $0 \leq nL \leq 255, 0 \leq nH \leq 255$

[Description] Sets the relative vertical print starting position from the current position in page mode.

This command sets the distance from the current position to $[(nL + nH \times 256) \times \text{vertical or horizontal motion unit}]$.

[Details]

- This command is ignored unless page mode is selected.
- When pitch N is specified to the movement downward:
 $nL + nH \times 256 = N$
- When pitch N is specified to the movement upward (the negative direction), use the complement of 65536.
When pitch N is specified to the movement upward:
 $nL + nH \times 256 = 65536 - N$
- Any setting that exceeds the specified printing area is ignored.
- This command function as follows, depending on the print starting position set by ESC T: When the starting position is set to the upper left or lower right of the printing, the vertical motion unit (y) is used.
- When the starting position is set to the upper right or lower left of the printing area, the horizontal motion unit (x) is used.
- The horizontal and vertical motion unit are specified by GS P.
- The GS P command can change the horizontal (and vertical) motion unit. However, the value cannot be less than the minimum horizontal movement amount, and it must be in even units of the minimum horizontal movement amount.

[Reference] ESC \$, ESC T, ESC W, ESC \, GS \$, GS P

2.57. GS ^ r t m (*)

[Name] Execute macro

[Format] ASCII	GS	^	r	t	m
Hex	1D	5E	r	t	m
Decimal	29	94	r	t	m

[Range] $0 \leq r \leq 255$. $0 \leq t \leq 255$. $m = 0, 1$.

[Description] Executes a macro.

- r specifies the number of times to execute the macro.
- t specifies the waiting time for executing the macro.
- m specifies macro executing mode. When the LSB of m = 0:

The macro executes r times continuously at the interval specified by t. When the LSB of m = 1:

After waiting for the period specified by t, the PAPER OUT LED indicators blink and the printer waits for the FEED button to be pressed. After the button is pressed, the printer executes the macro once. The printer repeats the operation r times.

[Details]

- The waiting time is $t \times 100$ ms for every macro execution.
- If this command is received while a macro is being defined, the macro definition is aborted and the definition is cleared.
- If the macro is not defined or if r is 0, nothing is executed.
When the macro is executed (m = 1), paper always cannot be fed by using the FEED button.

[Reference] GS :

2.58. GS a n (*)

[Name] Enable/Disable Automatic Status Back (ASB)

[Format] ASCII	GS	a	n
Hex	1D	61	n
Decimal	29	97	n

[Range] $0 \leq n \leq 255$

[Description] Enables or disables ASB and specifies the status items to include, using n as follows:

Table 2-27 The first byte

Bit	Off/On	Hex	Decimal	Status for ABS
0-1	Off	00	0	Not used Fixed to off
2	On	04	4	Drawer kick-out connector pin 3 is LOW
3	Off	00	0	online
4	On	10	16	Not used. Fixed to On.
5	Off	00	0	Cover is closed.
6	Off	00	0	Paper is not being fed by using the PAPER FEED button.
	On	40	64	Paper is being fed by using the PAPER FEED button.
7	Off	00	0	Not used. Fixed to off

Table 2-28 The second byte (printer information)

Bit	Off/On	Hex	Decimal	Status for ABS
0-4	Off	00	0	Undefined
5	Off	00	0	No unrecoverable error.
	On	20	32	Unrecoverable error occurred.
6	Off	00	0	No automatically recoverable error.
	On	40	64	Automatically recoverable error occurred.
7	Off	00	0	Not used. Fixed to Off.

Table 2-29 The third byte (paper sensor information)

Bit	Off/On	Hex	Decimal	Status for ABS
0,1	Off	00	0	Not used. Fixed to Off.
2,3	Off	00	0	Paper roll near-end sensor: paper near end.
	On	0C	12	Paper roll end sensor: paper present.
4-7	Off	00	0	Not used. Fixed to Off.

Table 2-30 The fourth byte (paper sensor information)

Bit	Off/On	Hex	Decimal	Status for ABS
0-3	-	-	-	Not defined.
4-7	Off	00	0	Not used. Fixed to Off.

2.59. GS f n (*)

[Name] Select font for Human Readable Interpretation (HRI) characters

[Format] ASCII GS f n
Hex 1D 66 n
Decimal 29 102 n

[Range] n = 0, 1, 48, 49

[Description] Selects a font for the HRI characters used when printing a bar code. n selects a font from the following table:

Table 2-31 n for ASCII characters type

n	Typeface
0,48	Standard ASCII characters (12 × 24)
1,49	Compressed ASCII characters (9 × 17)

[Details] HRI indicates Human Readable Interpretation.

HRI characters are printed at the position specified by GS H.

[Default] n = 0

[Reference] GS H, GS k

2.60. GS h n

[Name] Select bar code height

[Format] ASCII GS h n
Hex 1D 68 n
Decimal 29 104 n

[Range] $1 \leq n \leq 255$

[Description] Selects the height of the bar code.

n specifies the number of dots in the vertical direction.

[Default] n = 162

[Reference] GS k

2.61. ①GS k m d1...dk NUL ②GS k m n d1..dn

[Name] Print bar code

[Format] ① ASCII GS k m d1...dk NUL
Hex 1D 6B m d1...dk 00
Decimal 29 107 m d1...dk 0
② ASCII GS k m n d1...dn
Hex 1D 6B m n d1...dn
Decimal 29 107 m n d1...dn

[Range] ① $0 \leq m \leq 6$ (k and d depends on the bar code system used)

② $65 \leq m \leq 73$ (n and d depends on the bar code system used)

[Description] Selects a bar code system and prints the bar code.

m selects a bar code system as follows:

Table 2-32 ① Bar Code System Selection

m	Bar Code System	Number of characters	Character	Remarks	
①	0	UPC-A	$11 \leq k \leq 12$	0~9	$48 \leq d \leq 57$
	1	UPC-E	$11 \leq k \leq 12$	0~9	$48 \leq d \leq 57$
	2	JAN13 (EAN13)	$12 \leq k \leq 13$	0~9	$48 \leq d \leq 57$
	3	JAN8 (EAN8)	$7 \leq k \leq 8$	0~9	$48 \leq d \leq 57$
	4	CODE39	$1 \leq k \leq 255$	0~9,	$45 \leq d \leq 57,$
				A~Z	$65 \leq d \leq 90,$
				SP, \$, %, +,	d = 32, 36, 37,
				-, ., /	43, 45, 46, 47
				*(Start / end character)	d = 42(Start / end character)
	5	ITF	$1 \leq k \leq 255$ (Even number)	0~9	$48 \leq d \leq 57$
	6	CODABAR	$1 \leq k \leq 255$	0~9,	$48 \leq d \leq 57,$
				A~D	$65 \leq d \leq 68,$
				\$, +, -, .,	d = 36, 43, 45, 46,
				/, :,	47, 58

Table 2-33 ② Bar Code System Selection

m		Bar Code System	Number of characters	Character	Remarks
②	65	UPC-A	$11 \leq n \leq 12$	0~9	$48 \leq d \leq 57$
	66	UPC-E	$11 \leq n \leq 12$	0~9	$48 \leq d \leq 57$
	67	JAN13 (EAN13)	$12 \leq n \leq 13$	0~9	$48 \leq d \leq 57$
	68	JAN8 (EAN8)	$7 \leq n \leq 8$	0~9	$48 \leq d \leq 57$
	69	CODE39	$1 \leq k \leq 255$	0~9,	$45 \leq d \leq 57,$
				A~Z	$65 \leq d \leq 90,$
				SP, \$, %, +,	$d = 32, 36, 37,$
				-, ., /	43, 45, 46, 47
				*(Start / end character)	$d = 42(\text{Start / end character})$
	70	ITF	$1 \leq n \leq 255$ (Even number)	0~9	$48 \leq d \leq 57$
	71	CODABAR	$1 \leq n \leq 255$	0~9,	$48 \leq d \leq 57,$
				A~D	$65 \leq d \leq 68,$
				\$, +, -, .,	$d = 36, 43, 45, 46,$
				/, ;	47, 58
	72	CODE93	$1 \leq n \leq 255$	NUL~SP(7FH)	$0 \leq d \leq 127$
	73	CODE128	$2 \leq n \leq 255$	NUL~SP(7FH)	$0 \leq d \leq 127$

[Details for ①]

- This command ends with a NUL code.
- When the bar code system used is UPC-A or UPC-E, the printer prints the bar code data after receiving 12 bytes bar code data and processes the following data as normal data.
- When the bar code system used is JAN13 (EAN13), the printer prints the bar code after receiving 13 bytes bar code data and processes the following data as normal data.
- When the bar code system used is JAN8 (EAN8), the printer prints the bar code after receiving 8 bytes bar code data and processes the following data as normal data.
- The number of the data for ITF bar code must be even numbers. When an odd number of data is input, the printer ignores the last received data.

[Details for ②]

- n indicates the number of bar code data, and the printer processes n bytes from the next character data as bar code data.

- If n is outside of the specified range, the printer stops command processing and processes the following data as normal data. [Details in standard mode].
- If d is outside of the specified range, the printer only feeds paper and processes the following data as normal data.
- If the horizontal size exceeds printing area, the printer only feeds the paper.
- This command feeds as much paper as is required to print the bar code, regardless of the line spacing specified by ESC 2 or ESC 3.
- This command is enabled only when no data exists in the print buffer. When data exists in the print buffer, the printer processes the data following m as normal data.
- After printing bar code, this command sets the print position to the beginning of the line.
- This command is not affected by print modes (emphasized, double-strike, underline, character size, white/black reverse printing, or 90° rotated character, etc.), except for upside-down printing mode.

[Details in page mode]

- This command develops bar code data in the print buffer, but does not print it. After processing bar code data, this command moves the print position to the right side dot of the bar code.
- If d is out of the specified range, the printer stops command processing and processes the following data as normal data. In this case the data buffer position does not change.
- If bar code width exceeds the printing area, the printer does not print the bar code but moves the data buffer position to the left side out of the printing area.

When CODE128 (m = 73) is used:

Table 2-34 CODE128

Special character	Transmit data		
	ASCII	Hex	Decimal
SHIFT	{S	7B, 53	123, 83
CODE A	{A	7B, 41	123, 65
CODE B	{B	7B, 42	123, 66
CODE C	{C	7B, 43	123, 67
FNC1	{1	7B, 31	123, 49
FNC2	{2	7B, 32	123, 50
FNC3	{3	7B, 33	123, 51
FNC4	{4	7B, 34	123, 52
"{"	{{	7B, 7B	123, 123

- The printer prints an HRI character () as start character at the beginning of the HRI character string.
- The printer prints an HRI character () as a stop character at the end of the HRI character string.
- The printer prints HRI characters (F + an alphabetic character) as a control character (<00>H to <1F>H and <7F>H):

[Examples] example print "No. 123456"

In this example, the printer first prints "No." With CODE B, followed by the rest of the digital print CODE C:

GSK731012366781114612367123456



Figure 2-13 CODE C Example

- If the character is not in the forefront of barcode data set selection, the printer will stop processing this command, and the rest of the data as normal data.
- If "{" and followed by its combination of character is not specified above that, the printer stops processing this comm- and, and the rest of the data as normal data.
- If the printer is not receiving character code character set data, the printer stops processing this command, and the rest of the data as normal data.
- When the printer to print HRI characters do not print shift characters and character set selection data. HRI characters • function character is not printed.
- Control characters (<00> H to <1F> H and <7F> H) of HRI characters are not printed;
<Other> must ensure clearance around barcode. Gap varies by type of bar code.

[Reference] GS H, GS f, GS h, GS w

2.62. GS r n (*)

[Name] Transmit status

[Format] ASCII GS r n
 Hex 1D 72 n
 Decimal 29 114 n

[Range] n = 1, 2, 49, 50

[Description] Transmits the status specified by n as follows:

Table 2-35 Transmit Status

n	Function
0,48	Transmits paper sensor status
1,49	Transmits drawer kick-out connector status

[Details]

- This command just takes effect on the serial interface.
- This command is executed after the data in the receive buffer is processed. Therefore, there may be a time lag between receiving this command and transmitting the status, depending on the receive buffer status.
- The status types to be transmitted are shown below:

Table 2-36 Paper sensor status (n = 1, 49)

Bit	Off/On	Hex	Decimal	Status for ABS
0-1	Off	00	0	Paper roll near-end sensor: paper adequate.
	On	03	3	Paper roll near-end sensor: paper almost used up.
2-3	Off	00	0	Paper roll end sensor: paper adequate.
	On	0C	12	Paper roll end sensor: no paper.
4	Off	00	0	Not used. Fixed to Off.
5,6		—	—	Undefined.
7	Off	00	0	Not used. Fixed to Off.

Table 2-37 Cash drawer status (n = 2, 50)

Bit	Off/On	Hex	Decimal	Status for ABS
0	Off	00	0	Cash drawer opening.
	On	01	1	No cash drawer opening.
1-3		—	—	Undefined.
4	Off	00	0	Not used. Fixed to Off.
5,6		—	—	Undefined.
7	Off	00	0	Not used. Fixed to Off.

[Reference] DLE EOT, GS a

2.63. GS v 0 m xL xH yL yH d1...dk

[Name] Print raster bit image

[Format] ASCII GS v 0 m xL xH yL yH d1 ... dk

Hex 1D 76 30 m xL xH yL yH d1 ... dk

Decimal 29 118 48 m xL xH yL yH d1 ... dk

[Range] $0 \leq m \leq 3$, $48 \leq m \leq 51$

$0 \leq xL \leq 255$

$0 \leq xH \leq 255$

$0 \leq yL \leq 255$

$0 \leq d \leq 255$

$k = (xL + xH \times 256) \times (yL + yH \times 256) (k \neq 0)$

[Description] Selects Raster bit-image mode. The value of m selects the mode, as follows:

Table 2-38 Mode for Different Density

n	Mode	Vertical Dot Density	Horizontal Dot Density
0,48	Normal	200 DPI	200 DPI
1,49	Double-width	200 DPI	100 DPI
2,50	Double-height	100 DPI	200 DPI
3,51	Quadruple	100 DPI	100 DPI

- xL, xH, select the number of data bits (xL+ xH×256) in the horizontal direction for the bit image.
- yL, yH, select the number of data bits (yL+ yH×256) in the vertical direction for the bit image.

[Details]

- In standard mode, this command is effective only when there is no data in the print buffer.
- This command has no effect in all print modes (character size, emphasized, double-strike, upside-down, underline, white/black reverse printing, etc.) for raster bit image.
- If the printing area width set by GS L and GS W is less than the minimum width, the printing area is extended to the minimum width only on the line in question. The minimum width means 1 dot in normal (m=0, 48) and double-height (m=2, 50), 2 dots in double-width (m=1, 49) and quadruple (m=3, 51) modes.
- Data outside the printing area is read in and discarded on a dot-by-dot basis.
- The position at which subsequent characters are to be printed for raster bit image is specified by HT (Horizontal Tab), ESC \$ (Set absolute print position), ESC \ (Set relative print position), and GS L (Set left margin). If the position at which subsequent characters are to be printed is not a multiple of 8, print speed may decline.

Hex 1C 21 n

Decimal 28 33 n

[Range] $0 \leq n \leq 255$

[Description] Sets the print mode for Kanji characters, using n as follows:

Table 2-40 Print Mode for Kanji Characters

Bit	Off/On	Hex	Decimal	Function
0-1	—	—	—	Undefined.
2	Off	00	0	Double-width mode is OFF.
	On	04	4	Double-width mode is ON.
3	Off	00	.	Double-height mode is OFF.
	On	08	8	Double-height mode is ON.
4-6	—	—	—	Undefined.
7	Off	00	0	Underline mode is OFF.
	On	80	128	Underline mode is ON.

[Details]

- When both double-width and double-height modes are set (including right- and left-side character spacing), quadruple- size characters are printed.
- The printer can underline all characters (including right- and left-side character spacing), but cannot underline the space set by HT and 90° clockwise-rotated characters.
- The thickness of the underline is that specified by FS, regardless of the character size.
- When some of the characters in a line are double or more height, all the characters on the line are aligned at the baseline.
- It is possible to emphasize the Kanji character using FS W or GS !, the setting of the last received command is effective.
- It is possible to turn under line mode on or off using FS , and the setting of the last received command is effective.

[Default] n = 0

[Reference] FS -, FS W, GS !

2.66. FS &

[Name] Select Kanji character mode

[Format] ASCII FS &

Hex 1C 26

Decimal 28 38

[Description] Selects Kanji character mode.

[Reference] FS .,

2.67. FS - n

[Name] Turn underline mode on/off for Kanji characters

[Format] ASCII FS - n

Hex 1C 2D n

Decimal 28 45 n

[Range] $0 \leq n \leq 2, 48 \leq n \leq 50$

[Description] Turns underline mode for Kanji characters on or off, based on the following values of n.

Table 2-41 Underline mode for Kanji Characters

n	Function
0,48	Turns off underline mode for Kanji characters
1,49	Turns on underline mode for Kanji characters (1-dot thick)
2,50	Turns on underline mode for Kanji characters (2-dot thick)

[Details]

- The printer can underline all characters (including right- and left-side character spacing), but cannot underline the space set by HT and 90° clockwise-rotated characters.
- After the underline mode for Kanji characters is turned off by setting n to 0, underline printing is no longer performed, but the previously specified underline thickness is not changed. The default underline thickness is 1 dot.
- The specified line thickness does not change even when the character size changes.
- It is possible to turn underline mode on or off using FS !, and the last received command is effective.

[Default] n = 0

[Reference] FS !

2.68. FS .

[Name] Cancel Kanji character mode

[Format] ASCII FS .

Hex 1C 2E

Decimal 28 46

[Description] Cancels Kanji character mode.

[Reference] FS & ,

2.69. FS 2 c1 c2 d1...dk

Define user-defined characters

[Format] ASCII FS 2 c1 c2 d1 ... dk

Hex 1C 32 c1 c2 d1 ... dk

Decimal 28 50 c1 c2 d1 ... dk

[Range] c1, c2 define the character of the representative character encoding $C1=FEH, A1H \leq c2 \leq FEH, 0 \leq d \leq 255, k = 72$

[Description] Defined by the designated c1, c2 characters

[Default] c1, c2 on behalf of a user-defined character encoding, c1 specifies the first byte, c2 specify the second byte.

[Notes]

- d represents the data. 1 to print a dot, 0 means no printing point.
- No custom characters.

2.70. FS S n1 n2

[Name] Set left- and right-side Kanji character spacing

[Format] ASCII FS S n1 n2

Hex 1C 53 n1 n2

Decimal 28 83 n1 n2

[Range] $0 \leq n1 \leq 255, 0 \leq n2 \leq 255$

[Description]

- Sets left- and right-side Kanji character spacing n1 and n2, respectively.
- When the printer model used supports GS P, the left-side character spacing is $[n1 \times \text{horizontal or vertical motion units}]$, and the right-side character spacing is $[n2 \times \text{horizontal or vertical motion units}]$.

[Details]

- When double-width mode is set, the left- and right-side character spacing is twice the normal value.
- The horizontal and vertical motion units are set by GS P. The previously specified character spacing does not change, even if the horizontal or vertical motion unit is changed using GS P.
- The value cannot be less than the minimum horizontal movement amount, and must be in even units of the minimum horizontal movement amount.
- In standard mode, the horizontal motion unit is used.
- In page mode, the horizontal or vertical motion unit differs in page mode, depending on starting position of the printable area as follows:

① When the starting position is set to the upper left or lower right of the printable area using ESC T, the horizontal

motion unit (x) is used.

② When the starting position is set to the upper right or lower left of the printable area using ESC T, the vertical motion unit (y) is used.

③ The maximum right-side spacing is 255/180 inches for the paper roll and is approximately 35.983 mm {255/150 inches}. Any setting exceeding the maximum is converted to the maximum automatically.

[Default] n1 = 0, n2 = 0

[Reference] GS P

2.71. FS W n

[Name] Turn quadruple-size mode on/off for Kanji characters

[Format] ASCII FS W n

Hex 1C 57 n

Decimal 28 87 n

[Range] $0 \leq n \leq 255$

[Description] Turns quadruple-size mode on or off for Kanji characters.

- When the LSB of n is 0, quadruple-size mode for Kanji characters is turned off.
- When the LSB of n is 1, quadruple-size mode for Kanji characters is turned on.

[Details]

- Only the lowest bit of n is valid.
- In quadruple-size mode, the printer prints the same size characters as when double-width and double-height modes are both turned on.
- When quadruple-size mode is turned off using this command, the following characters are printed in normal size.

- When some of the characters on a line are different in height, all the characters on the line are aligned at the baseline.
- FS ! or GS ! can also select and cancel quadruple-size mode by selecting double-height and double-width modes, and the setting of the last received command is effective.
- [Default] n = 0 [Reference] FS !, GS !
- The printer features a command prompt

2.72. ESC B n t

[Name] Printer to print single Beeper

[Format]	ASCII	ESC	B	n	t
	Hex	1B	42	n	t
	Decimal	27	66	n	t

[Range] 1 <= n <= 9, 1 <= t <= 9

[Description] printer to print single Beeper.

- n is the number of times the buzzer.
- t is the number of tweets buzzer each time (t × 50) ms

2.73. ESC C m t n

[Name] Printer to print single Beeper and alarm light flashes

[Format]	ASCII	ESC	C	m	t	n
	Hex	1B	43	m	t	n
	Decimal	27	67	m	t	n

[Range] 1 <= m <= 20, 1 <= t <= 20, 0 <= n <= 3,

[Description]

- printer to print single Beeper and alarm lights blinking.
- m: 1 <= m <= 20, refers to the warning light flashes the number of times or buzzer.
- t: 1 <= t <= 20, refers to the warning lights flashing at intervals of t * 50 ms or buzzer interval of (t × 50) ms.
- When n = 0, the buzzer does not beep while warning light does not blink;
- When n = 1, the buzzer sounds; when n = 2, warning lights flashing;
- When n = 3, the buzzer sounds, while warning lights flashing;

3.0. 128 Code

3.1. 128 Yard Review

128 yards through the alternate character sets A, B character set and character set C, capable of 128 and 100 ASCII characters from 00 to 99 digits, and some special characters are encoded. Each character set encoding of characters is as follows:

Character Set A: ASCII character codes 00H to 5FH

Character Set B: ASCII character codes 20H to 7FH

Character Set C: 00 ~ 99 of 100 digital

128 yards on the following special characters can be encoded:

SHIFT character "SHIFT" to make the character behind the bar code symbol SHIFT first character from the character set A character set conversion to B, or from character set to character set B A, began to recover from the second character to the previously used SHIFT character set. "SHIFT" character set only character set conversion between A and B used in the character, it does not make the current coded character set C character enters or exits the state.

Character set selection character (CODE A, CODE B, CODE C) These characters can convert it back to a character set encoding of characters A, B or C.

function character (FNC1, FNC2, FNC3, FNC4) the usefulness of these features characters depends on the application software. In the character set C, only FNC1 available.

3.2. Character set

3.2.1. A Set of Characters

Table 3-1 A Set of Characters

Characters	Send data		Characters	Send data		Characters	Send data	
	Hex	Decimal		Hex	Decimal		Hex	Decimal
NUL	00	0	(	28	40	P	50	80
L	01	1	)	29	41	Q	51	81
SOH	02	2	*	2A	42	R	52	82
STX	03	3	+	2B	43	S	53	83
ETX	04	4	,	2C	44	T	54	84
EOT	05	5	-	2D	45	U	55	85
ENQ	06	6	.	2E	46	V	56	86
ACK	07	7	/	2F	47	W	57	87
BEL	08	8	0	30	48	X	58	88
BS	09	9	1	31	49	Y	59	89
HT	0A	10	2	32	50	Z	5A	90
LF	0B	11	3	33	51	[	5B	91
VT	0C	12	4	34	52	\	5C	92
FF	0D	13	5	35	53	]	5D	93
CR	0E	14	6	36	54	^	5E	94
SO	0F	15	7	37	55	-	5F	95
SI	10	16	8	38	56	FNC	7B,31	123,49
DLE	11	17	9	39	57	1	7B,32	123,50
DC1	12	18	:	3A	58	FNC	7B,33	123,51
DC2	13	19	;	3B	59	2	7B,34	123,52
DC3	14	20	<	3C	60	FNC	7B,53	123,83
DC4	15	21	=	3D	61	3	7B,42	123,66
NAK	16	22	>	3E	62	FNC	7B,43	123,67
SYN	17	23	?	3F	63	4		
ETB	18	24	@	40	64	SHIF		
CAN	19	25	A	41	65	T		
EM	1A	26	B	42	66	COD		
SUB	1B	27	C	43	67	EB		
ESC	1C	28		44	68	COD		

FS	1D	29	D	45	69	EC		
GS	1E	30	E	46	70			
RS	1F	31	F	47	71			
US	20	32	G	48	72			
SP	21	33	H	49	73			
!	22	34	I J	4A	74			
"	23	35	K	4B	75			
*	24	36	L	4C	76			
\$	25	37	M	4D	77			
%	26	38	N	4E	78			
&	27	39	O	4F	79			

3.2.2. B Set of Characters

Table 3-2 A Set of Characters

Characters	Send data		Characters	Send data		Characters	Send data	
	Hex	Decimal		Hex	Decimal		Hex	Decimal
SP	20	32	H	48	72	p	70	112
!	21	33	I	49	73	Q	71	113
"	22	34	J	4A	74	r	72	114
*	23	35	K	4B	75	S	73	115
\$	24	36	L	4C	76	t	74	116
%	25	37	M	4D	77	u	75	117
&	26	38	N	4E	78	v	76	118
'	27	39	O	4F	79	W	77	119
(	28	40	P	50	80	X	78	120
)	29	41	Q	51	81	Y	79	121
*	2A	42	R	52	82	z	7A	122
+	2B	43	S	53	83	{	7B,7B	123,123
,	2C	44	T	54	84		7C	124
-	2D	45	U	55	85	}	7D	125
.	2E	46	V	56	86	—	7E	126
/	2F	47	W	57	87	DEL	7F	127
0	30	48	X	58	88	FNC	7B,31	123,49
1	31	49	Y	59	89	1	7B,32	123,50
2	32	50	Z	5A	90	FNC	7B,33	123,51
3	33	51	[	5B	91	2	7B,34	123,52
4	34	52	\	5C	92	FNC	7B,53	123,83
5	35	53	]	5D	93	3	7B,41	123,65
6	36	54	^	5E	94	FNC	7B,43	123,67
7	37	55	`	5F	95	4		
8	38	56	~	60	96	SHIF		
9	39	57	a	61	97	T		
:	3A	58	b	62	98	COD		
;	3B	59	c	63	99	EA		
<	3C	60	d	64	100	COD		
=	3D	61	e	65	101	EC		
>	3E	62	f	66	102			
?	3F	63	g	67	103			
@	40	64	h	68	104			
A	41	65	i	69	105			
B	42	66	j	6A	106			
C	43	67	k	6B	107			
D	44	68	l	6C	108			
E	45	69	m	6D	109			
F	46	70	n	6E	110			
G	47	71	o	6F	111			

3.2.3. C Set of Characters

Table 3-3 A Set of Characters

Characters	Send data		Characters	Send data		Characters	Send data	
	Hex	Decimal		Hex	Decimal		Hex	Decimal
0	00	0	40	28	40	80	50	80
1	01	1	41	29	41	81	51	81
2	02	2	42	2A	42	82	52	82
3	03	3	43	2B	43	83	53	83
4	04	4	44	2C	44	84	54	84
5	05	5	45	2D	45	85	55	85
6	06	6	46	2E	46	86	56	86
7	07	7	47	2F	47	87	57	87
8	08	8	48	30	48	88	58	88
9	09	9	49	31	49	89	59	89
10	0A	10	50	32	50	90	5A	90
11	0B	11	51	33	51	91	5B	91
12	0C	12	52	34	52	92	5C	92
13	0D	13	53	35	53	93	5D	93
14	0E	14	54	36	54	94	5E	94
15	0F	15	55	37	55	95	5F	95
16	10	16	56	38	56	96	60	96
17	11	17	57	39	57	97	61	97
18	12	18	58	3A	58	98	62	98
19	13	19	59	3B	59	99	63	99
20	14	20	60	3C	60	FNC 1	7B,31	123,49
21	15	21	61	3D	61	COD	7B,41	123,65
22	16	22	62	3E	62	EA	7B,42	123,66
23	17	23	63	3F	63	COD		
24	18	24	64	40	64	EB		
25	19	25	65	41	65			
26	1A	26	66	42	66			
27	1B	27	67	43	67			
28	1C	28	68	44	68			
29	1D	29	69	45	69			
30	1E	30	70	46	70			
31	1F	31	71	47	71			
32	20	32	72	48	72			
33	21	33	73	49	73			
34	22	34	74	4A	74			
35	23	35	75	4B	75			
36	24	36	76	4C	76			
37	25	37	77	4D	77			
38	26	38	78	4E	78			
39	27	39	79	4F	79			

4.0. Print Mode and Its Conversion

Summary printer has two operating modes: standard mode and page mode. In standard mode, as long as the line printer buffer is full, or received print or feed command, the printer will print the paper in hand. In page mode, all of the print data and feed commands are in the specified memory space, the printer does not perform any operation is stored. Until it receives an ESC FF or FF command, the printer will print all the content areas are printed.

For example: When the printer is in standard mode received "ABCDEF" <LF>, it immediately "ABCDEF" to print out a line of paper in hand. In page mode, the printer will "ABCDEF" is written in memory of the print area, the next print data on the next line of the print area.

4.1. Print Mode Switching

ESC L command to switch the printer to the page mode, all data are processed after this in page mode. All data can thus execute ESC FF command received print out all the data, but can only execute FF command received print out, and the printer will switch to standard mode. ESC S command execution can also switch the printer to standard mode, but it does not print in page mode the received data, and these data will be cleared.

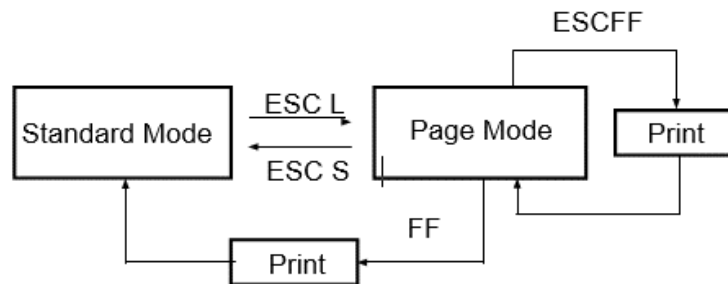


Figure 4-1 Print Mode Switch

4.2. Set Various Values

Some commands (eg: ESC SP, ESC 2, ESC 3, and FS S) can be used both in standard mode and can be used in page mode, and their arguments are the same. However, provided in two modes are independent, they are stored, respectively.

4.3. Set Print Area

Print area is set by ESC W command. If the ESC W command before receiving all of the print and paper feed operations have been completed, the printer to the left (as you face the printer) as the print area of the origin of coordinates (x0, y0). Width (dx dot) extension of the rectangular printing area to the right in the x- direction (perpendicular to the paper feed direction) from the origin of coordinates (x0, y0); Height (dy point) in the y direction (paper feed direction). If you do not use ESC W set the print area, the print area using default values.

After the printer in the print area and the print area direction (by the ESC T command set) is set, the received print data will be arranged according to the position within the print area, A point for the starting position of the print area this is a default value. (When a character is printed, A point as the baseline) print data downloaded bitmap or barcode data to its current position as the lower left corner, aligned with the baseline.

Upon receiving a command containing feed (eg: Before LF or ESC J), if the print data (including character spacing) has exceeded the print area, the printer automatically feeds the line (how much feed, determined by the ESC line-height 2 and ESC 3 settings), while the printing position to the next column header.

The default line height is 1/6 inches, equivalent to 31 points on the vertical. If the line containing the next print data in the longitudinal upscaling 2 times the characters, or bitmap occupies two lines or more lines, as well as higher than normal character code, the printer cannot meet the needs of the feed volume, resulting in a character on a line of characters printed and the printed overlay. To avoid this, the line can be increased higher.

For example,

When printing a height of 6 bytes downloaded bitmap using the formula below:

{Vertical dots (8×6) - printing area starting position feed point (24)} $\times$ vertical motion unit ($200 / 200$) = 24, that is, to print the bitmap is complete, will require the print position based on the starting position of the area down 24 points. With the following commands:

ESC W xL, xH, yL, yH, dxL, dxH, dyL, dyH

ESC T n

ESC 3 24 Å to set a new line of high-

LF

GS / 1

ESC 2 Å The row height back to the 1/6 inches.



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